

Adversarial Attacks on Deep Learning-based Network Intrusion Detection Systems: A Taxonomy and Review

ABSTRACT Network Intrusion Detection Systems (NIDS) protect crucial network and data resources by serving as the first line of defense against cyber threats. However, using deep learning (DL) in NIDS makes them highly vulnerable to adversarial attacks that affect their reliability. This paper presents a taxonomy and review of DL-based adversarial attacks and investigates the application of these classified attacks in the context of NIDS. We categorised attacks based on model knowledge, training-testing attack, invisibility metric, attack goal, and focus area. Furthermore, we explore DL-based adversarial attack defenses within NIDS, present evaluation metrics, benchmark datasets, and adversarial tools, and analyze existing commercial intrusion detection tools. Our analysis provides insights into deep learning-based NIDS approaches and emphasizes the challenges in strengthening their robustness against evolving cyberattacks. The study also highlights the need for future research to enhance NIDS's ability to prevent adversarial attacks and improve network security.

INDEX TERMS Adversarial Attack, NIDS, NIDS Tool, Machine Learning, Deep Learning, Benchmark Dataset

I. INTRODUCTION

Over the past few decades, there has been remarkable progress in the evolution of computer networks. Innovations in network topology and protocols have significantly enhanced communication among devices across various scenarios, which are not confined to tiny networks, i.e., the Internet of Things (IoT). Researchers are also developing large complex networks [1]. The growing number of network devices brings an increased vulnerability to many security threats which can compromise the confidentiality, integrity, and availability of critical assets, such as data, systems, and whole networks [2].

Many network attacks have been initiated including password cracking, IoT malware, and man-in-the-middle attacks, and DoS attacks present significant challenges to the security of networks. Considering these threats, intrusion detection systems (IDS) have been a key security solution development. IDS are responsible for detecting and countering potential network attacks, ensuring the system is protected from harmful activities. IDS are classified into network-based IDS (NIDS) and Host-based IDS (HIDS), depending on their deployment location. In contrast to HIDS which is focused on monitoring a single device [3], NIDS performs network-wide surveillance, thus providing a holistic approach to threat detection.

The deployment of NIDS typically involves three phases: monitoring, detection, and response [4]. The monitoring phase collects statistical features related to the network state including the number of sent packets and connections made. In the detection phase, Machine Learning (ML) algorithms process these attributes to spot potential intrusion attempts. In the last step, the system answers with appropriate measures to neutralize detected threats.

In recent years, a significant change has occurred in the machine learning techniques used for anomaly-based NIDS design, where deep learning (DL) models now dominate these systems [5]. Owing to the rapid growth of the volume of network traffic, conventional ML algorithms become less effective, and the ability of DL algorithms to fully utilize the massive dataset draws attention. The appeal of DL in network security is its ability to learn expressive and non-linear representations, enabling the classification of network threats with a very high accuracy [6].

This advancement has not been obtained without problems because DL-based methods are vulnerable to adversarial attacks small noises in input can result in misclassification. Researchers are interested in different methods to prevent adversarial attacks, specifically in the critical security domain of Network Intrusion Detection Systems (NIDS). The need for secure and resilient networks grows with time, and incor-

porating advanced technologies is crucial to defend against constantly evolving threats and maintain the health of digital ecosystems [7].

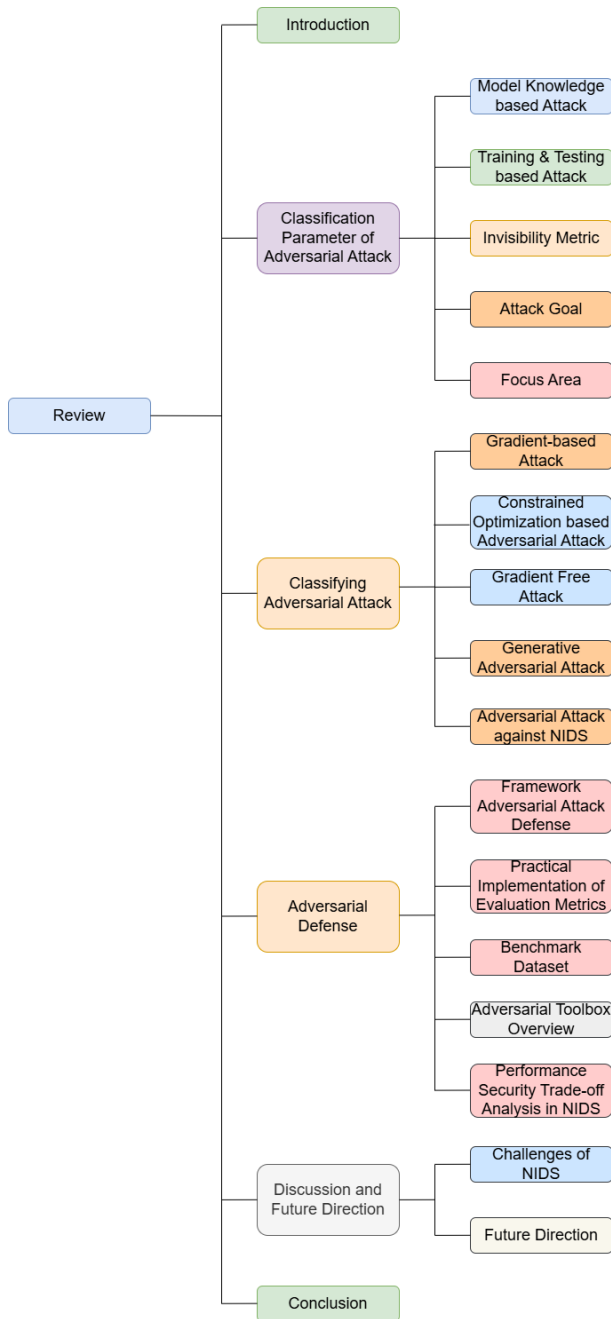


FIGURE 1. The anatomy of this review

Related Survey: In recent years, IDS has seen great advancements, especially anomaly-based NIDS. Researchers actively develop high-level traffic classification and outlier detection algorithms, such as algorithms based mainly on ML and DL techniques. The integration of DL algorithms, especially, has gained a large following in the NIDS arena because they can leverage large datasets and generate highly expressive features which are employed in accurate threat

detection.

Previous surveys, for instance, the study of Buczak and Guven [8], developed thorough taxonomies of NIDS and examined ML and DL techniques applied to anomaly-based detection. Chaabouni et al. [9] scrutinized ML algorithms purposely developed for NIDS in IoT environments focusing on unique challenges constituted by IoT devices. Hussain et al. [10] extended their scope to IoT security in general, testing ML and DL approaches for addressing a wide spectrum of IoT security problems such as malware analysis, authentication, access control, and intrusion detection. Zhang et al. [11] explored ML and DL algorithms for security and analytical tasks in wireless and mobile networks, adapting DL algorithms to suit different mobile network types. Although these surveys examined ML/DL algorithms in detail within their respective domains, they failed to tackle adversarial attacks and defense mechanisms that DL algorithms are specifically exposed.

The adversarial vulnerability of deep neural networks (DNNs) has become an urgent issue. Most existing works on adversarial attacks and defenses are centered around image classifiers [12]. Wang et al. [13] presented a taxonomy of attacks and defenses classified by the stage of the training process targeted by the attack, with a focus on data privacy protection during the training. The authors of [14] performed a mathematical analysis of the significant white-box (WB) and black-box (BB) adversarial attacks and corresponding strengths of the training as well as at test time methods to remove the adversarial perturbations. Chakraborty et al. [15] implemented a wider scope of the definition of adversarial attacks that include exploratory, poisoning, and evasion attacks, and indicated respective defense strategies. Silva and Najafirad [16] considered WB and BB attacks, emphasizing methods that improve the robustness of networks, while Serban et al. [17] discussed adversarial attacks and defenses in the context of computer vision tasks. Bhambri et al. [18] focused on BB threat model attacks and defenses in computer vision tasks. Zhang et al. [19] examined adversarial attacks and defenses in the image domain and proposed to adapt these attacks to textual inputs for natural language processing (NLP) tasks.

Despite the comprehensive surveys, investigating adversarial attacks on DL algorithms in cybersecurity, especially in the context of NIDS, is still a little-discovered field. Olowononi et al. [20] offered a general taxonomy of ML/DL algorithms dealing with the Cyber Physical System (CPS), then considered adversarial attacks and defenses in CPS. Martins and co-authors [21] introduced noteworthy WB and BB attacks against ML/DL approaches related to NIDS and malware detection. Ibitoye et al. [22] reviewed ML and DL algorithms used for network security tasks, as well as adversarial attacks and defense techniques. Alatwi and Morisset [23] studied on the influence of adversarial examples on ML-based NIDS, exploring current attacks and defenses in NIDS.

Unfortunately, the previous works have focused on providing methods for adversarial attacks and defenses without giv-

ing a unified and abstract view of the topic. Additionally, they have not sufficiently considered the feasibility of adapting adversarial attacks designed for images to the NIDS domain as traffic and image features have substantial differences. On the other hand, none of the studies has mentioned any classification of adversarial attacks in the field of NIDS. This also suggests an important research gap in studying adversarial attacks and defenses relating to NIDS. This study addresses this gap as an attempt to present an elaborate and easy-to-follow survey of adversarial attacks and defenses against NIDS, which reveals the practical implications of the attacks and closes the gap between the image-oriented studies and the specific characteristics of network traffic data.

The comprehensive survey we conducted aims to bridge the knowledge gaps regarding adversarial attacks and defenses in Network Intrusion Detection Systems (NIDS). The primary contributions of this survey can be outlined as follows:

- 1) Classify deep learning-based adversarial attacks and investigate the application of these classified attacks in Network Intrusion Detection Systems (NIDS);
- 2) Examination of DL-based Adversarial Attack Defenses within NIDS;
- 3) Presentation of evaluation metrics, benchmark datasets, and adversarial tools for NIDS;
- 4) Analysis of existing Commercial Intrusion Detection Tools.

II. CLASSIFICATION PARAMETER OF ADVERSARIAL ATTACK

A. MODEL KNOWLEDGE-BASED ATTACK

Knowledge-based attacks, in the realm of cybersecurity, usually revolve around exploiting vulnerabilities by utilizing insights into a target system. This may encompass comprehending the software, network structure, or weaknesses within the system. Adversarial attacks can be classified into three main categories based on the level of information the attacker has about the target model: white-box attacks, black-box attacks, and gray-box attacks. **White-box Attack:** A white-box attack [24] is where an attacker knows either the decision boundary for a linear classifier or all weight values and structure for a deep neural network of the targeted system. After obtaining specific knowledge, the attacker has fixed a goal to create adversarial examples to satisfy confidence and similarity constraints. As a result, the attacker can bypass or manipulate the input feature of the targeted system.

Black-box Attack: A black-box [25], [26] is the opposite of white-box attacks. Black-box attacks are hidden attacks on machine learning models, where the adversary knows nothing about the model's internal architecture, the neural network's weights, and the training data. The adversary creates adversarial examples to fool the model by performing querying of the model with input data and noting the outputs of the predictions. The most popular techniques to make advisory samples in a black-box setting include: transfer-based attacks, query-based attacks, and decision-based attacks.

Gray-box Attack: A gray-box attack [27] is where an attacker has partial knowledge about the targeted system. For instance, the attacker does not know the classifier but has preprocessing knowledge.

B. TRAINING AND TESTING-BASED ATTACK

Based on training and testing data, adversarial attacks can happen in two ways, evasion attacks and poisoning attacks. Evasion is always known as an exploratory attack or inference time attack. On the other hand, a poisoning attack is also recognized as a causative attack or training time attack.

Poisoning Attack: To misclassify an object, the attacker puts false training data in the target system. There are multiple ways to inject training data into the targeted system. For instance, during binary classification, an attacker may simply flip labels [28]. Similarly, data is put on the wrong cluster while the targeted system adopts unsupervised classification [29].

Evasion Attack: Evasion is a kind of attack where an attacker wants to bypass the trained model of a target system. Adversarial examples are specially formulated to be nearly indistinguishable from the true data points in a dataset but pose threats where they manipulate flaws inherent within model decision limiters [30]. These instances are usually derived from deliberately controlled perturbations in this input data. However, the attacker has access to the model's architecture and parameters that enable it to generate inputs that are not visible to the human eye but cause incorrect outputs from this model. Generally, an evasion attack is used to bypass spam detectors or network intrusion systems.

C. INVISIBILITY METRIC

Perturbation: Perturbation is noise or additional data added to the original data to create an adversarial example. A distance vector is a numerical representation of the difference between two data points each feature space. In the case of adversarial attacks, the distance vector measures the difference between the original input and the perturbed input. The most commonly used distance metrics for adversarial perturbations are the l_p norms, such as l_0, l_1 (Manhattan distance), l_2 (Euclidean distance), and l_∞ (maximum absolute difference). A distance matrix is a matrix in the form of a square that has the pairwise distances among the set of data points. In the case of adversarial attacks, the distance matrix helps understand the relationship between multiple perturbed inputs and the original one. Each (i, j) element in the distance matrix signifies the distance between the i-th and j-th data points. Distance matrices are especially popular when comparing the performance of several adversarial attack methods or comparing the resistance of a model to different interferences.

If r is the distance vector and a is the length vector, then the norm of a will be denoted as $\|a\|$. Then, the distance matrix l_p will be defined For a given vector $\mathbf{a} = (a_1, a_2, \dots, a_n)$, the

p -norm is defined as:

$$\|\mathbf{a}\|_p = \left(\sum_{i=1}^n |a_i|^p \right)^{\frac{1}{p}}$$

where $p \geq 1$. If $p = 1$, the distance matrix is given by:

$$\|\mathbf{a}\|_1 = \sum_{i=1}^n |a_i|$$

For the distance function between the original data sample $\mathbf{a}$ and the adversarial data sample $\mathbf{a}_0$ with $p = 1$, the l_0 -norm is a convex surrogate of the l_0 -norm:

$$f(\mathbf{a}) = \|\mathbf{a} - \mathbf{a}_0\|_1$$

If $p = 2$, the Euclidean norm is defined as:

$$\|\mathbf{a}\|_2 = \left(\sum_{i=1}^n |a_i|^2 \right)^{\frac{1}{2}}$$

The distance function between the original and adversarial samples with $p = 2$ is the l_2 -norm:

$$f(\mathbf{a}) = \|\mathbf{a} - \mathbf{a}_0\|_2$$

For $p = \infty$, the distance function is:

$$f(\mathbf{a}) = \|\mathbf{a} - \mathbf{a}_0\|_\infty$$

and the l_∞ -norm is given by:

$$\|\mathbf{a}\|_\infty = \max_{i=1, \dots, n} |a_i|$$

which means the perturb element will be minimized. The selection of perturbation is a major factor related to the success of adversarial attacks. Small (imperceptible) imperfections are preferred because they are less likely to be detected and their presence does not ruin the original content [31]. Nevertheless, the efficiency of these small-scale changes will depend on the attack algorithm type and the target model's resistance. Various perturbation methods are commonly used like the Fast Gradient Sign Method (FGSM) which uses the sign of the gradient to determine the perturbation direction and CW attack [32] that optimizes the perturbation to achieve the successful misclassification with minimum magnitude of the perturbation. Both the attack success rate and the imperceptibility of the chosen perturbation method and its magnitude vary depending on the specific needs and constraints of the adversarial attack environment, and therefore it differs from one scenario to another.

D. ATTACK GOAL

Based on the attack goal, there are two types of objectives: targeted or untargeted. If the original example class is C_0 and another class is C_k , where a_0 is the original data point and $a_0 \in \mathbb{R}^d$, then the main goal of the adversarial attack is to modify the original data so that the model is classified as C_k instead of C_0 . If the attacker aims to use specific perturbations for the system/model to classify a specific class

C_k , it is known as a targeted attack. On the other hand, if the attacker aims to use any class C_k (not C_0), it is known as an untargeted attack.

Targeted Attack: If the perturbation vector is δ , then the adversarial example is defined as $a = a_0 + \delta$, where $\delta \in \mathbb{R}^d$. Considering a multiclass classifier with classes $C_0, C_1, C_2, \dots, C_K$ and discriminant functions $g_i(a)$ for $i = 1, 2, 3, \dots, K$, the adversarial attack will be a targeted attack if it satisfies $g_t(a) \geq g_j(a)$ for all $j \neq t$, and the constraint will be $\max_{j \neq t} \{g_j(a)\} - g_t(a) \leq 0$.

Untargeted Attack: An adversarial attack will be considered as an untargeted attack if it satisfies $g_i(a) \leq g_j(a)$ for all $j \neq i$. Similarly, the constraint will be $\bar{g}_i(a) - \min_{j \neq i} \{g_j(a)\} \leq 0$.

E. FOCUS AREA

Adversarial attacks pose significant threats in diverse domains such as Networks and Communications [33], Computer Vision [34], Autonomous Driving, Finance, e-Commerce, and Human-machine Interactions, etc. Choosing a focus area to classify adversarial attacks is vital because not all attacks provide significant outcomes for targeted models to misclassify objects.

Network Communications: As network communications adopt machine learning and deep learning to secure data and privacy, adversarial attacks have a tremendous impact and exploit vulnerabilities. Studies on wireless communication reveal that adversarial attacks significantly increase the bit error rate during data transmission in wireless mediums [35]. Adversarial attack-generating algorithms like FGSM and Projected Gradient Decent (PGD) are now used for human identification data, authentication data, and face recognition, etc., exploiting cybersecurity vulnerabilities [36], [37]. For example, a slight perturbation is enough to break down fingerprint data, with a relatively high success rate [38].

Computer Vision: Computer vision, dealing mainly with image data, is vulnerable to adversarial attacks. Attackers use perturbations to create adversarial attacks and successfully bypass or make trained models misclassify. For instance, tiny changes in input data using adversarial perturbations confuse face recognition systems, resulting in misidentifications [34]. Similarly, slight changes in input data lead to significant misclassifications in object detection [39]. Laser beams on road signs can cause trained models to fail to detect correct labels [40], exemplifying challenges in the physical world. Additionally, adversarial attacks in medical imagery have been introduced and studied extensively [41].

Unlike network communication and computer vision, adversarial attacks significantly impact areas such as robotics, finance, e-commerce, and more. In robotics, dependent on sensor data like voices and images, attackers can easily inject adversarial perturbations to cause misclassification [42]. In the financial sector, despite adopting machine learning and deep learning, adversarial examples pose serious issues, leading to detection systems being bypass [43]. Purchasing and trade systems behave incorrectly due to adversarial examples,

where tiny changes in input data are sufficient to mislead the trained system [44]. Bank credit scores are also affected by adversarial examples, allowing attackers to easily change credit scores with minor alterations in input data [45].

III. CLASSIFYING ADVERSARIAL ATTACKS

An adversarial attack is one such kind of malicious attempt that shifts the features of the original object to another object, resulting in the misclassification of the original object. If the original feature vector is denoted as $\mathbf{a}_0$ (where $\mathbf{a}_0 \in \mathbb{R}^d$) and belongs to class C_i , and another object's feature vector $\mathbf{a}$ (where $\mathbf{a} \in \mathbb{R}^d$) belongs to class C_t , the adversarial attack mapping $\chi : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is defined as $\mathbf{a} = \chi(\mathbf{a}_0) = \mathbf{a}_0 + \mathbf{r}$. To generate an adversarial example by adding perturbation to the original feature, the adversarial data will be $\mathbf{a} = \chi(\mathbf{a}_0) = \mathbf{a}_0 + \mathbf{r}$, where $\mathbf{r}$ is an additional feature used to create the perturbation.

Based on the above theory, multiple ways exist to generate examples of adversarial attacks. Many studies have been conducted for creating adversarial attacks and each attack has considered several factors according to attack goal circumstance. our studies have shown about classification of adversarial attacks by considering some classification parameters that are already in Section 2. The classification of adversarial attacks is crucial for identifying an architecture defense mechanism.

A. GRADIENT-BASED ATTACKS

Gradient-based approaches are a common method for creating adversarial examples, particularly designed for deep neural networks. This strategy relies on the gradient of input data increasing the loss function. Several gradient-based attacks have been explored. In the following paragraphs, we discuss some of these methods conducted by various researchers.

FGSM: Goodfellow et al. (2014) [31] introduced the Fast Gradient Signed Method (FGSM) attack, aiming to maximize the loss function $J(a, w)$, where a is the feature vector and w is the model's parameters (weights and biases). The perturbation process is bounded by the l_∞ -norm, expressed mathematically as:

$$a = a_0 + \mu \cdot \text{sign}(\nabla_a J(a_0; w))$$

Here, the added perturbation is given by $r = \mu \cdot \text{sign}(\nabla_a J(a_0; w))$, where μ is a constant used to control the gradient descent. If FGSM is applied for linear classification with the the trained model on the original example is $O(a) = w_i^T a + w_{i0}$ and the targeted adversarial targeted function $t(a) = w_t^T a + w_{t0}$, then the loss function of FGSM is defined as $J(a; w) = -(w^T a + w_0)$. Here, the optimizing loss function $J(a; w)$ will result from an FGSM attack.

However, FGSM possesses certain weaknesses in respect to sample generation of adversarial instances. FGSM does not bring out the optimal perturbations every time, and the method might not get the most useful adversarial samples. Further on, FGSM is a single-step algorithm, which means

that with a shift in the direction of the gradient it only takes one step to get the adversarial example.

I-FGSM: To address issues when applying FGSM under bounded conditions, Kurakin et al. (2017) [25] proposed the Iterative Fast Gradient Sign Method (I-FGSM). It generates iterative input data using the equation:

$$a^{(k+1)} = \arg \max_{0 \leq a \leq 1} \nabla_a J(a^k; w)^T a$$

The adversarial attack is denoted as:

$$a^{(k+1)} = \mathcal{P}_{[0,1]} \{a^k + \mu \cdot \text{sign}(\nabla_a J(a^k; w))\}$$

Here, $\|a - a^k\|_\infty \leq \mu$, and k represents the number of iterations and $\mathcal{P}_{[0,1]}$ is a projection operator that elementwisely projects out of bound values to bound $0 \leq a \leq 1$

MI-FGSM: To improve black-box attack efficiency, Li et al. (2018) [29] proposed Momentum Iterative FGSM (MI-FGSM), defined for the l_∞ -norm as:

$$a_{k+1} = a_k - \tau \cdot \text{sign}(z_{k+1})$$

Here, τ is a small step size, and $z_{k+1} = \psi \cdot z_k + \frac{\nabla_a J(a; w)}{\|\Delta_a J(a; w)\|_1}$, where ψ is the decay factor.

BIM: Building on FGSM, [46] demonstrated a straightforward modification by applying a clip to pixel values and a small step size for gradient descent. BIM is formulated as:

$$a^{(k+1)} = \text{Clip}_a(\mu) \{a^k + \tau \cdot \text{sign}(\nabla_a J(a^k; w))\}$$

here, $\text{Clip}_{(\mu)} = \min\{255, a + \mu, \max\{0, a - \mu, a_0\}\}$.

Carlini-Wagner Attack (C&W): Some gradient-based attacks have issues when applied in unbounded conditions. Carlini-Wagner (C&W) is a gradient-based attack that addresses this unboundedness issue. Carlini-Wagner et al. (2016) [47] introduced a regulation-based attack. If the discriminant function is denoted as $g(a)$, and the objective function is $\phi(a, k) = \|a - a_0\|^2 + \lambda \max\{\max_{j \neq t} g_j(a) - g_t(a), 0\}$, then the Carlini-Wagner attack is defined as:

$$a^{(k+1)} = a^k - \tau \nabla \phi(a^k; i^*)$$

Here i^* is the parametrizing factor, and the gradient depends on how i^* is defined. So, $i^* = \arg \max\{g_j(a^k)\}$.

JSMA: JSMA [47] is another gradient-based attack, where the gradient is applied from the hidden layer to the output. Saliency maps serve as informative guides, illustrating the specific input features that an adversary should strategically perturb to efficiently achieve the desired alterations in network output. Consequently, these versatile tools empower adversaries to generate various adversarial samples. If the second to the last layer of the neural network is denoted as $L(a) = L_1(a), L_2(a), L_3(a), \dots, L_k(a)$, then the saliency mapping will be

$$S[a, t][i] = \begin{cases} 0, \nabla_a L_t(a) < 0 \text{ or } \sum_{k \neq t} \nabla_a L_k(a) > 0 \\ \nabla_a L_t(a) \mid \sum_{k \neq t} \nabla_a L_k(a), \text{ otherwise} \end{cases}$$

TABLE 1. Abbreviation

Abbreviation	Full form	Abbreviation	Full form
ACAC	Average Confidence of Adversarial Class	KL	Kullback Leibler divergence
ACGAN	Adversarial Sample Detector	KNN	k-Nearest Neighbour
AdGAN	Adversarial GAN	L-BFGS	Limited-memory BFGS
Advcam	Adversarial Camouflage	LAFeAT	Latent FEATure Attack
AdvDrop	Adversarial Drop	LDA	Latent Dirichlet Allocation
AdvLB	Adversarial Laser Beam	LF-Low	Frequency component distortion
AECGAN	Auto-Encoding CGAN	LPIPS	Learned Perceptual Image Patch Similarity
AE	Annotated GAN	LR	Logit Reconstructor
ALAD	Apple Leaf Disease Detection	LSTM	Long Short Term Memory
ALT-GAN	Adversarial Learned Inference GAN	LTMU	long-term tracker with a meta-updater
AMP	Average Modification Percentage	Mal-IGAN	Malware Image Analysis GAN
ANN	Artificial Neural Network	MANDA	MANifold and Decision boundary-based AE detection system
AoA	Attack on Attention	MAP	Multispectral Adversarial Patch
ARIES GAN	smArt gRid Intrusion dEtection	MES	Mobile Edge Computing
ASR	Attack Success Rate	MI-FGSM	Momentum Fast Gradient Sign Method
Atkse	Attacking by Shrinking Errors	MIM	Man in Middle Attack
ATTA	Adversarial Transformation-enhanced Transfer Attack	ML	Machine Learning
BB	Black Box	MLP	Multi-Layer Perception
BCAS	Bit Coordinate Ascent	MultiD-WGAN	Multi-Discriminator Wasserstein GAN
BGAS	Multi-Step Bit Gradient Ascent	NB	Naïve Bayes
BiGAN	Bidirectional GAN	NC	Network Communication
Bi-LSTM	Bi-directional LSTM	NIDS	Network Intrusion Detection System
BIM	Basic Iterative Method	NLP	Natural Language Processing
BN	Bayesian Network	NSGA-PSO	Non-dominated Sorting Genetic Algorithm with Particle Swarm Optimization
CAT	Contrastive Adversarial Training	NSS	Normalized Scan-path Saliency
C-LSTM	CNN with Long Short-Term Memory	Opt Attack	Optimization Attack
CNN	Convolution Neural Network	PCA	Principal Component Analysis
CoR-GAN	Correlation GAN	PCAE	Principal Component Analysis
CPS	Cyber Physical Attack	PGD	Projected Gradient Decent
CV	Computer Vision	PLC	Pearson's Linear Coefficient (minus)
CW	Carlini and Wagne attacking algorithm	PS-GAN	Perceptual-Sensitive GAN
DAC	Degree Assortative Change	PW	Physical World
DAGMM	Deep Auto-encoding Gaussian Mixture Model	PWPSA	Probability Weighted Packet Saliency Attack
DCGAN	Deep Convolutional Generation	QDA	Qualitative Data Analysis
DEF-IDS	Defence-Intrusion Detection System	RF	Random Forest
dFGSM	Deterministic Approach FGSM	RFGAN	Random Forest Support Vector Machine
DL	Deep Learning	rFGSM	Randomize Rounding FGSM
DNN	Deep Neural Network	RMSE	Root Mean Squared Error
DSNGD	Dynamically Sampled Nonlocal Gradient Descent	RNN	Recurrent Neural Network
DSVDD	Deep Support Vector Data Description	RNN-ADV	Random Neural Network for Adversarial
DT	Decision Tree	RNN-AdV	Random Neural Network Adversarial
DW	Digital World	ROC	Receiver Operating Characteristic Curve
FaceAdv	Face Adversarial	RP2	Robust Physical Perturbation
FeaCP	Feature-wise Convex Polytype attack	SCA	Side Channel Attack
FGSM	Fast Gradient Sign Method	SDWSN	Software Defined Wireless Sensor Network
FID	Frechet Inception Distance	SGA	Simplified Gradient-based Attack
FNN	Feed-forward Neural Network	SGADV	Similarity-based Gray-box Adversarial Attack
GAN	Generative Adversarial Network	SGAN	Self-Attention GAN
GB	Gradient boosting	SIGSM	Selective and Iterative Gradient Sign Method
GCNN	Gated Convolutional Neural Network	SNN	Spiking Neural Network
GEM	Graph Embedding Models	SRLIM	Surrogate Representation Learning with Isometric Mapping
GF-Attack	Generalized adversarial attack Framework	SSAH	Semantic Similarity Attack on High-frequency components
GFT	Gradient Fusion Technology	SSIM	Structure Similarity
GIDS	GAN based Intrusion Detection System	SVM	Support Vector Machine
HAG	Hash Adversary Generation	UAP	Universal Adversarial Perturbation
HSJA	HopSkipJumpAttack	WB	White Box
IBA	Iterative Black-box Attack	WGAN	Wasserstein GAN
IDSGAN	Intrusion Detection System GAN	ZAQ	Zero-shot Adversarial Quantization
IGA	Iterative Gradient Attack		
I-FGSM	Iterative Fast Gradient Sign Method		
IoU	Intersection over Union		
IPW	Iterative Partially Saliency Map Attack		
JSMA	Jacobian-based Saliency Map Attack		

Here, $\nabla_a L_t(a)$ and $\sum_{k \neq t} \nabla_a L_k(a)$ identify how much $L_t(a)$ increases and $\sum_{k \neq t} L_k(a)$ decreases, given a modification of the input feature i . After getting the saliency mapping, the JSMA attack will be defined as

$$a^{(k+1)} = a^k - \tau \nabla \phi(x^k; i^*),$$

where $i^* = \arg \max \{S[a, t][i]\}$.

DeepFool: [48] proposed a gradient-based attack for computing robustness in deep classifiers, if $C(a)$ is the classifier

function that maps the input a to its class label, then the l_2 -norm

$$a^{(k+1)} = a^k - \frac{C(a)}{\|\nabla_a C(a)\|_2^2} \nabla_a C(a)$$

PGD: The PGD (Projected Gradient Descent) adversarial attack, introduced by Madry et al. [49], is a powerful “first-order” type of onslaught. The gradient projection devoted algorithm (PGD) itself is a well-known optimization method

that projects gradients onto the specified ball. In essence, PGD is a version of iterative FGSM with a small step size, which highly resembles the Basic Iterative Method (BIM) [50]. Interestingly, PGD starts randomly inside the ball and involves random restarts. The iterative process of PGD is illustrated by the following equation

$$a^{(k+1)} = \prod_{(a+S)} (a^k + \tau \cdot \text{sign}(\nabla_a L(\theta, a, a_0)))$$

In this case, the symbol $L(\cdot)$, stands for the loss function where x denotes the input to the model described by parameters denoted θ , and a corresponds to the target of a_0 . The expression $a+S$ represents the projection operator associated with the perturbation set $x+S$, while τ denotes the step size of the gradient. The feasible set S is usually defined as an l_p norm ball: $S = a : \|a - a_0\|_p$

where a_0 is the original input, is the maximum allowed perturbation, and p is the type of norm used (e.g., $p = 2$ for Euclidean distance, $p = \infty$ for maximum absolute difference). It is important to emphasize that this study uses the cross-entropy loss function.

DNNs are one of those tools that have been widely used in solving network graph-related problems involving node discrimination and link prediction. Among them, an advanced solution in this field is the Iterative Gradient Attack (IGA) [55], which has a new perspective on link prediction. IGA utilizes the gradient data acquired from a pre-trained GAE model. It is interesting to note that IGA displays efficiency in scenarios of both complete and partial information graphs indicating its ability to do several tasks. Secondly, IGA has outstanding transferability between various diagrammatic representations across different practical contexts. However, it should be noted that the computational complexity of IGA may grow exponentially concerning a growing graph size.

IPW [56] is a new entry in adversarial attacks introduced by Che et al. This technique plans the cost function in a critical hidden space marked by high receptive field activity. Compared to conventional techniques, IPW punishes certain feature activations connected with both defining and guiding areas rather than penalizing every pixel spread over the whole dense output area. Furthermore, Che et al. propose an IBA [56] which uses non-redundant variations from base models as starting hints to measure the gradients of a target black box model. The estimation process follows a zeroth-order iterative optimization strategy, wherein directional derivatives are computed along non-redundant initial directions.

DSNGD: This adversarial attack is motivated by a PGD attack and adopts a Zero-order attack [57] as a baseline. This attack considers that attack can able to know only a few features which is known as partial white box or gray box. In the hidden layer, one of the parts is considered a bottleneck, and calculate the loss function of the bottleneck. The main motivation of this attack is that if a tiny fraction changes in a hidden layer which gains massive values in the input layer non-salient image will tend to zero. Chen et al. [61] proposed the Attack on Attention (AoA) method that

uses shared semantic characteristics among multiple DNNs for the transferability of adversarial attacks. AoA differs from other methods by modifying attention heat maps and performs better on black-box attacks. There are significant transferability improvements when using attention loss as a substitute CE.

Li et al. [63] proposed an SGA for large-scale graph node classification tasks based on the Gradient Fusion Technique (GFT). SGA accomplishes this by strategically utilizing a smaller subgraph centered around the target node, which efficiently addresses gradient fading issues and surpasses both GradArgmax and Nettack methods regarding time efficiency.

SGA has been demonstrated to be effective in large-scale graph node classification, but when faced with malicious nodes that are singletons and unlinked, it becomes difficult for the k-hop subgraph extraction which causes huge computational cost. Further studies will focus on making SGA more effective for various graph analysis tasks. Wang et al. [60] proposed Surrogate Representation Learning with Isometric Mapping (SRLIM), which solves the problem of model-specific gradient-based perturbations in graph-based attacks by increasing topological knowledge transfer from input to embedding layers. Although SRLIM has improved effectiveness, its computational complexity increases as the graph structure becomes more complicated.

Xiang et al. [59] proposed a gray-box attack based on SCA, which uses the target model's network structure obtained through SCA and trains an alternative model producing adversarial samples; while being in between white-box attacks and black-box ones, this method proves significantly computationally demanding because of many candidate architectures to be evaluated.

Liu et al. [62] proposed the AtkSE attack model to overcome vulnerabilities in GNNs by introducing semantic invariance and momentum gradient integration modules that aim at reducing errors deriving from graph structure discreteness, as well as unstable representations of GNN nodes due to locality/semantic space or random parameter initialization; however, computational efficiency can be traded against accuracy improvement.

B. CONSTRAINED OPTIMIZATION-BASED ADVERSARIAL ATTACK

Constrained optimization-based attacks constitute an approach for delineating the spectrum between maximum and minimum allowable norm attacks. In the context of minimum allowable attacks, characterized by perturbation data (denoted as a), original data (a_0), and a discriminant function ($g(a)$), the optimization objective is formulated as follows:

$$\min_a \|a - a_0\| \quad \text{s.t.} \quad \max_{j \neq t} \{g_j(a)\} - g_t(a) \leq 0$$

Conversely, for the definition of maximum allowable attacks, the optimization problem takes the following form:

$$\min_a \max_{j \neq t} \{g_j(a)\} - g_t(a) \quad \text{s.t.} \quad \|a - a_0\| \leq \epsilon$$

TABLE 2. Gradient-based Adversarial Attack Parameter Analysis

Attack	Input	KB	Invis- ibility Metric	Year	Testing and Training Data	Attack Goal	Appli- cation	Focus Area	Used IDS	Ref
FGSM	Continuous	BB, WB	l_∞	2015	Evasion, poisoning	Targeted, untargeted	DW	NC, CV	Yes	[31], [26], [30], [28], [51]
I-FGSM	Continuous	BB, WB	l_∞	2015	Evasion	Targeted	DW	NC, CV	Yes	[25], [24], [52]
MI-FGSM	Continuous	BB, WB	l_∞	2015	Evasion	Targeted, untargeted	DW	NC, CV	Yes	[29]
BIM	Continuous	BB, WB	l_∞	2015	Evasion	Targeted, untargeted	DW	NC, CV	Yes	[50]
CW	Continuous	BB, WB	l_∞	2015	Evasion	Targeted, untargeted	DW	NC, CV	Yes	[32]
JSMA	Continuous	BB, WB	l_∞	2016	Evasion	Targeted, untargeted	DW	NC, CV	Yes	[53]
Deep Fool	Continuous	BB, WB	l_∞	2016	Evasion	Targeted, untargeted	DW	NC, CV	Yes	[54]
PGD	Continuous	BB, WB	l_∞	2018	Evasion	Targeted, untargeted	DW	NC, CV	Yes	[37]
IGA	Continuous	BB, WB	Cross Entropy	2020	Evasion	Targeted, untargeted	PW	CV	No	[55]
IPW	Continuous	BB, WB	KL, PLC, NSS, l_∞	2021	Evasion	Targeted, untargeted	DW	CV	No	[56]
IBA	Continuous	BB	KL, PLC, NSS, l_∞	2021	Evasion	Targeted, untargeted	DW	NC, CV	Yes	[56]
DSNGD	Continuous	BB, WB	l_∞	2021	Evasion	Targeted, untargeted	DW	CV	No	[57]
LAFEAT	Continuous	BB, WB	l_∞	2021	Evasion	Targeted, untargeted	DW	CV	No	[58]
SCA-based	Continuous	BB, WB	l_∞	2021	Evasion	Targeted, untargeted	DW	CV	No	[59]
SRLIM	Continuous	BB, WB	l_0	2021	Poisoning	Targeted, untargeted	DW	CV	No	[60]
AoA	Continuous	BB, WB	RMSE	2022	Evasion	Targeted, untargeted	DW	CV	No	[61]
AtkSE	Continuous	BB, WB	l_0	2022	Evasion	Targeted, untargeted	DW	CV	No	[62]
SGA	Continuous	BB, WB	DAC	2023	Evasion	Targeted, untargeted	DW	CV	No	[63]

where $\epsilon > 0$ serves as the optimization factor, representing the magnitude of the attack. This framework provides a nuanced exploration of attack strategies by considering both the minimal and maximal norm constraints within the optimization process.

An L-BFGS algorithm was employed by Szegedy et al. [64] to transform the difficult problem of finding perceptually minimal input perturbations into a box-constrained formulation, with many other methods such as CW [32], AdvGAN [65] and UAP also using bounded optimization strategies in terms of adversarial attacks.

NSGA PSO [66], a method based on optimization shows success rates in creating perturbations for digital watermarking, compared to existing black box attack methods. It demonstrates transferability across network models and improved resistance, against countermeasures aimed at image modification. However, its performance is relatively lower when applied to the CIFAR 10 dataset as compared to ImageNet. ATTA [67] employs learning to train a CNN as a transformation network improving its resilience and ability to transfer knowledge by capturing harmful deformations. However, the performance of ATTA is impacted by finding the balance, between simplicity and complexity in the network structure. There is potential for enhancing its effectiveness through integration, with transfer-based attack tech-

niques. In their study [68], Luo and colleagues presented the Semantic Similarity Attack on High-Frequency Components (SSAH), which uses space limitations to confine noise to the high-frequency elements of an image. This approach aims to reduce similarity while improving resemblance. However, the authors also noted a trade-off between invisibility and the success rate of the attack, posing a challenge, for creating attacks.

Chang and colleagues introduced GF Attack [69], a framework for attacks on graph embedding models (GEMs) that works without access to labels or predictions. The goal of this framework is to improve the resilience of GEMs. Although GF Attacks may not be as computationally efficient as other methods, they showcase transferability, flexibility, and adaptability across different types of GEMs without modifying the target embedding model.

C. GRADIENT-FREE ADVERSARIAL ATTACK

Gradient-free attacks, which encompass search-based, decision-based, and drop-based methods, are a type of attack independent of gradients in machine learning.

The paper [71] presents FeaCP, a method for constructing interpretable adversarial input samples around decision boundaries, utilizing feature-wise convex interpolation for adaptive neighborhood generation and expansion to adver-

TABLE 3. Constrained optimization-based Adversarial Attack Parameter Analysis

Attack	Input	KB	Invi-sibility Metric	Year	Test- ing and Training Data	Attack Goal	Appli- cation	Focus Area	Used IDS	Ref
adGAN	Discreate	WB	l_2	2021	Evasion	Targeted, untar- geted	DW	CV	No	[70]
NSGA-PSO	Continuous	WB,BB	l_2	2021	Evasion	Targeted, untar- geted	DW	CV	No	[66]
ATTA	Continuous	BB, WB	l_∞	2021	Evasion	Targeted, untar- geted	DW	CV	No	[67]
SSAH	Continuous	WB	FID, LF, ASR, l_∞	2022	Evasion	Targeted, untar- geted	DW	CV	No	[68]
GF-Attack	Continuous	BB	Accuracy, l_∞	2022	Evasion	Targeted, untar- geted	DW	CV	No	[69]

serial fine-tuning for the optimized decision boundaries.

The paper [72] introduces two novel physical-world adversarial attack methods: Adversarial Laser Beam (AdvLB), based on crafting malicious laser beams to deceive DNNs, and Adversarial Camouflage (AdvCam) [73], emphasizing the synthesis of inconspicuous and natural-style adversarial examples-together they widen the perception of real-world threats to deep learning models.

The paper [74] focuses on the deep hashing model's robustness concerning adversarial examples generated in Hamming space by introducing Hash Adversary Generation (HAG) to produce the imperceptible modifications that fool retrieval systems, highlighting the imperative of robustness when designing secure large-scale nearest neighbor search applications. Different from previous attacks that claim to produce undetectable examples, AdvDrop [75], a novel method of adversarial attacks, generates discreet examples by selectively removing image features, demonstrating robustness in the presence of existing defenses and offering an interesting way to evaluate the resilience of DNN, with the focus on frequency domain change.

Embedding a similarity-based gray-box method, SGADV [76] addresses the lack of label training data in face recognition authentication, exposing the vulnerability of the pre-trained models and highlighting the privacy risks, though less accurate than FGSM [31], PGD [77] and DeepFool [54], without investigating its transferability.

The IoU attack [78], a decision-based, black-box approach for visual object tracking, which exploits IoU scores from the current and past video frames, inserts particular changes that reduce the accuracy of temporally uniform bounding boxes.

The paper [76] addresses adversarial attacks on LSTM classifiers in intrusion detection systems, focusing on the importance of robustness in preserving network security, exploring the JSMA and FGSM attack methods, and proposing prevention techniques to ensure network safety.

The HopSkipJumpAttack [79] family of decision-based adversarial attack algorithms mainly focuses on generating adversarial examples by estimating the gradient direction using binary information at the decision boundary, showcasing efficiency by requiring fewer model queries than Boundary Attack and competitive performance against a wide range of defense mechanisms.

OptAttack [80] presents the adversarial attack problem as a continuous real-valued optimization problem and efficiently attacks the algorithm by using optimization algorithms in the zeroth order to find the perturbed adversarial direction and distance to the decision boundary.

This paper [81] proposes an improved genetic algorithm for adversarial attacks, which refines the crossover and mutation operations, is tested on 15 test functions, and outperforms state-of-the-art swarm intelligence optimization algorithms for neural network attacks without requiring internal model information.

D. GENERATIVE ADVERSARIAL NETWORK

As concerns on Network Intrusion Detection Systems (NIDS), generative adversarial attacks (GANs) are an attack mechanism that diverges from feature-based attacks. Generally, the GANs start by producing a random input that is fed to the discriminator to determine the difference between benign and malicious components. This procedure repeats until the discriminator fails to distinguish between benign or malicious.

In enhancing traditional Network IDS machine learning models, researchers leverage GANs such as DCGAN [85] and LSTM to create an effective real-time intrusion detection system that utilizes generated synthetic data for improved balance, achieving a high accuracy of 99.73% and 99.62% on KDD and NSL-KDD datasets. Another GAN-based system in a different study [86] attains a remarkable f1-score of over 94% for detecting malicious traffic by learning patterns in traffic log data.

A group of researchers in [87] experimented with the introduction of dual GAN schemes for network security, using one GAN to create adversarial attacks and another to defend against them. As the overall accuracy was not especially high, the study demonstrated the feasibility of GAN-based approaches for mitigating those attacks on machine learning or deep learning-based IDS models. This represents a creative, and two-party security model with a huge research potential.

The study [88] introduced a GAN-centered model that can become aware of opcodes from programs to detect malware variations, mostly in high-security systems. This method has resulted in a significant upgrade of detection precision from

TABLE 4. Gradient-free Adversarial Attack Parameter Analysis

Attack	Input	KB	Invisibility Metric	Year	Testing and Training Data	Attack Goal	Application	Focus Area	Used IDS	Ref
FeaCP	Continuous	WB	Error rate, l_∞	2021	Poisoning	Targeted, untargeted	PW	CV	No	[71]
AdvLB	Continuous	WB	Success Rate, l_∞	2021	Evasion	Targeted, untargeted	PW	CV	No	[72]
HAG	Continuous	WB	Hamming Distance	2020	Evasion, poisoning	Targeted, untargeted	DW	CV	No	[74]
Adv- Drop	Continuous	WB	Accuracy Rate, LPIPS	2021	Evasion	Targeted, untargeted	DW	CV	No	[75]
PCAE	Continuous	WB	Accuracy, l_∞	2020	Evasion	Targeted, untargeted	DW	CV	No	[82]
IoU	Continuous	WB	Accuracy, Cosine Distance	2021	Evasion	Targeted, untargeted	DW	CV	No	[78]
SGADV	Continuous	WB	Cosine Distance, ROC, SSIM, LPIPS, ASR	2021	Evasion	Targeted, untargeted	DW	CV	No	[76]
Hop- Skip-Jump	Continuous	BB, WB	l_∞	2022	Evasion	Targeted, untargeted	DW	CV	No	[79]
Bounded Attack	Continuous	BB, WB	l_∞	2022	Evasion	Targeted, untargeted	DW	CV	Yes	[83]
Opt Attack	Continuous	BB, WB	l_∞	2015	Evasion	Targeted, untargeted	DW	CV	No	[80]
PW- PSA	Discrete	BB	Success rate, AMP, Queries, ACAC	2021	Evasion	Targeted, untargeted	DW	NC	Yes	[84]
Genetic Attack	Discrete	BB	Success rate, AMP, Queries, ACAC	2021	Evasion	Targeted, untargeted	DW	NC	No	[81]

96.3% to 98%. When training data is augmented with the generated samples offered by the GAN, the gains are obvious even with meager data. Furthermore, in [89], the authors opened the Mal-IGAN model which applies GANs to detect malware samples in the form of images. It shows robustness with more than an 80% accuracy even while being trained with only 1% of the dataset, demonstrating the effectiveness of that model.

Replica [90] uses an LSTM GAN-based model for API call detection inside MS Windows executable files for detecting malicious codes. The suggested LSTM-Attention and BiLSTM-Attention classifiers based on contextual and semantic analysis of API calls deliver maximum accuracy of 95.43% and 96.53% with the help of machine learning models, respectively detecting malware samples.

The authors of [91] experiment with the GAN model which uses RGB images of malware as inputs and keeps the IDS learning tool up-to-date so that antivirus software systems can evolve over time. Such an algorithm, also known as the online learning approach, reinforces the "train and update" process described in [91]. DCGAN, ALI-GAN, CoR-GAN, and the augmented data set all consistently demonstrate good precision, recall, f1-score, and accuracy metrics in the cited paper.

In [92], Convolutional Neural Networks and WGANs are used together for a new IDS system with high accuracy and recognition of unseen attack samples. Additionally, [93] portrays the DoS-WGAN technique as a trace evidence generator for training IDS on machine learning-based defense systems, showing the attackers' ability to adapt to those defense systems. AI-driven NIDS through the GAN and Auto-encoder method shows a competitive performance across all

the datasets, and GAN and Random Forest algorithm secure great relevance in GAN using the networks as IDS owing to their performance in data recognition and the utility of GANs in creating efficient IDS.

A critical element of maintaining the security of the networked infrastructure is network intrusion detection systems (NIDS) that act at the network level. As proof of that, the solution FlowGAN [94] had strong results handling both encrypted and non-encrypted traffic. Also, the focus of another study [95] was on using the Conditional GAN Auto-encoder (AE-CGAN) to learn the most important features, to fit the IDS models with dynamic network environments. The importance of GANs in wireless network security and intrusion detection, is highlighted in a comprehensive 2022 review [96], giving access to all features and creating a culmination of adversarial examples for attacking and training purposes.

In [97], attackGAN draws attention to the fact that machine learning models are exposed to data perturbation on IoT devices thus signaling a call for the development of better strategies. A distributed identification system for IoT systems, proposed in [96], has a maximum of 20% higher accuracy than non-distributed methods and provides the networks with resistance to individual device failures. [98] utilizes BiGAN models in IoT dataset-23 for detecting Mirai botnet-like attacks and shows the F1-score of 99%, while [99] employs Wasserstein GAN and Auto-encoder for the IDS of an IoT network and the high confidence and success rates in a quest to achieve real-world applications.

Considering the pervasive use of mobile devices and by stand-alone security issues, [92] applies Wasserstein GAN for mobile malware detection, the accuracy of which reaches 88% in detecting suspicious network behavior. This figure

risers to 96.89% when the model is fed with generated data. Additionally, it applies a GAN model and ZipNet techniques to show that the citywide scale traffic analysis could be provided with a high level of accuracy because the depth of the Normalized Root Mean Square Error was significant. This, that further research into securing mobile networks from intrusions and malicious traffic still must be done.

The authors of [100] specify the vital need to overcome vulnerabilities of sensor networks, such as those in Software-Defined Wireless Sensor Networks (SDWSNs), by integrating machine learning/AI techniques with cryptographic methods. The ARIES GAN system for intrusion detection in the Smart Grid networks [101] is presented attaining 80% accuracy to protect critical national infrastructure and, reflecting the need for research on secure IDS models, particularly because of the tremendous growth of smart sensors in critical systems.

The research in [102] focuses on security in autonomous vehicles (AV). It highlights the vulnerability of connected systems and the transition toward artificial intelligence and machine learning for AV security. [103] introduces GIDS which employs the GAN model to examine the potential role of neutral networks in facilitating strong IDS detection with real-time image analysis of a wide range of vehicle attacks.

In the paper [104], the authors suggested a security system based on the GAN scheme that will be deployed in each network edge node, especially for MEC servers. Here, monitoring is carried out on the traffic to and from the MEC server, the accuracy of which can reach 90%, showing the importance of GANs in the security of network devices in autonomous vehicles.

Alhajjar et al. [23] introduced more target classifiers and used Genetic Algorithm and Particle Swarm Optimization to generate adversarial data. Their main research focus is preserving network semantics by discriminating mutable and immutable features and evaluating adversarial examples using UNSW-NB15 and NSL-KDD datasets.

Kalbayer et al. [105] presented IDSGAN which leverages predefined domain constraints and a loss function that is more than just making the discriminator unable to distinguish adversarial from benign traffic, but also minimizes the distance between generated and original features to achieve more evasive attacks.

TABLE 5. Taxonomy of Generative Adversarial Network

Focus Area	Ref
Network Intrusion Detection System	[85], [87], [88], [89], [106], [90], [91], [92], [93], [107], [108], [109], [110], [111], [112]
Wireless Network	[96], [94], [95], [113]
Internet of Things	[114], [115], [99], [100], [101], [116], [117], [118], [119], [120], [121], [122], [123], [124], [125]
Mobile Intrusion	[126], [127]
Sensor Network Intrusion	[128], [129], [130]
Autonomous Vehicle Intrusion	[102], [131], [104]

E. ADVERSARIAL ATTACK AGAINST NIDS

In this section, we discuss dedicated adversarial attacks intended to bypass and misclassify target systems within NIDS. The previous section briefly discussed different types of adversarial attacks based on their performance characteristics. Here, however, we focus on attacks specifically designed for NIDS, excluding others that might not be relevant in this context.

Feature-level attacks for Network Intrusion Detection Systems (NIDS) include gradient-based, constraint-based, and gradient-free attacks that directly manipulate network traffic features. Many studies have been done on feature-based attacks, mostly on gradient-based methods like FGSM, I-FGSM, MI-FGSM, L-BFGS, PGD, JSMA, Deepfool, CW, MC, and BIM. Also, constraint optimization attacks, including PSO attacks, are used in NIDS analysis. In addition, gradient-free attacks such as PWPSA and Boundary Attack are also employed in NIDS scenarios.

Adversarial attacks pose challenges across various machine learning algorithms, including MLP, SVM, DT, RF, NB, KNN, LSTM, GB, and others. A detailed summary of these attacks employed in Network Intrusion Detection Systems (NIDS) is presented in Table 5. In assessing the efficacy of these adversarial attacks, researchers utilize diverse benchmark datasets, such as NSL-KDD, Bot-IoT, IEEE-bigdata2019, CICIDS2017/18, UNSW-NB15, and more, as discussed in the subsequent sections.

Summary: The effectiveness of feature-level adversarial attacks has been shown by numerous experiments with various approaches such as gradient-based, constrained optimization-based, and gradient-free (shown in Tables 2 ,3,4). However, all of the adversarial attacks are not adopted by bypassing NIDS. In the table 6, we have shown that several feature-based attacks have been used for bypassing network intrusion detection system. Nevertheless, a major drawback of feature-level attacks is their impracticability, especially when working with structured datasets such as network traffic data. In real life when the characteristics of data are not interchangeable, performing simple perturbations is not applicable.

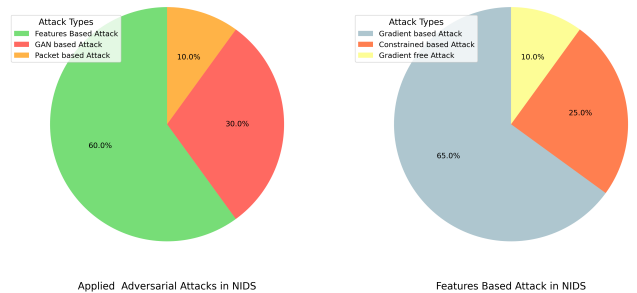


FIGURE 2. Adversarial attack application overview in NIDS

The practical limitation of this concept has been resolved with the introduction of generative adversarial networks

TABLE 6. Adversarial Attack against NIDS Parameter Analysis

Attack	Year	Bypass AI Model	KB	Dataset	Testing and Training Data	Ref
FGSM	2015	MLP, ANN, SVM, GAN, DNN, FNN, SNN, DT, RF	WB	NSL-KDD, Bot-IoT, IEEEbig-data2019Cup, CICIDS2017, CIC-DDoS2019, UNSB-NB15	Evasion, poisoning	[47], [48], [77]
I-FGSM	2015	DNN	WB	Kitsune	Evasion	[132]
MI-FGSM	2015	DNN,SVM, RFLR	WB	NSL-KDD	Evasion	[49]
L-BFGS	2015	DNN,SVM, RFLR	WB	NSL-KDD	Evasion	[49]
PGD	2015	DNN,SVM, RF,LR, FNN,SNN	WB	NSL-KDD,Bot-Iot	Evasion	[77], [49]
JSMA	2015	MLP,RF, SVM,DT	WB,GB	NSL-KDD,Bot-Iot,UNSB-NB15	Evasion	[133], [47], [48]
Deep- Fool	2015	MLP	WB	NSL-KDD	Evasion	[48]
CW	2015	MLP,FNN, SNN	WB	NSL-KDD, UNSB-NB15, Bot-IoT	Evasion	[47], [48]
MC	2015	SVM,DT, NB,KNN, RF,MLP, GB,LR, LDA,QDA, BAG	WB	NSL-KDD, UNSB-NB15	Evasion	[134]
GAN	2015	SVM,DT, NB,KNN, RF,MLP, GB,LR, LDA,QDA, BAG,MLP, KNN,DNN	WB,BB, GB	NSL-KDD, UNSB-NB15, NSKKDD, CICIDS2017	Evasion	[135], [136], [134]
PWPSA	2015	LSTM	BB	CICIDS2017	Evasion	[137]
BFAM	2015	LR,DT, MLP,NB, RF	BB	NSL-KDD	Evasion	[136]
PSO	2015	SVM,DT, NB,KNN, RF,MLP, GB,LR, LDA,QDA, BAG	WB	NSL-KDD, UNSB-NB15	Evasion	[134]
GA	2015	SVM,DT, NB,KNN, RF,MLP, GB,LR, LDA,QDA, BAG	WB	NSL-KDD, UNSB-NB15	Evasion	[134]
Boundary based Attack	2015	ANN	BB	KDDcup99, CICIDS2017	Evasion	[83]
SPSA	2015	Stratosphere Linux IPS, DNN, SVM, RF, LR	GB,WB	CTU-108-1, NSL-KDD	Evasion	[49]
BIM	2015	SVM, MLP, KNN, CNN, Resnet, FNN, SNN	BB, WB	KDDcup99, CICIDS2018, Bot-IoT	Evasion	[46]
Spherical Local sub-spaces	2015	DAGMM, AE, AnoGAN, ALAD, DSVDD,one-class, SVM, IF	BB	CICIDS2018	Evasion	[138]
Custom attack	2015	RF,MLP, KNN, Stratosphere, Linux IPS	BB,GB	CTU-13, CTU-108	Evasion, poisoning	[139], [140]
Re-labeling	2015	DNN	WB	NSL-KDD	poisoning	[141]
Gen-AAL	2015	DT	GB	CICIDS2017	Evasion	[142]
IDS- GAN	2018	MLP	WB	NSL-KDD	Evasion	[135]
ZOO	2018	DNN	GB	NSL-KDD	Evasion	[143]

(GANs) in the field of Network Intrusion Detection Systems (NIDS) (shown in Table 5). Although GANs have demonstrated promising outcomes in generating adversarial features, the major obstacle arises when moving from device to network levels, making them less related to real-world applications.

On the other hand, packet-based attacks [138], [144], [145] become a secondary choice for practical-based attacks [146], [147], [148]. However, not much research has been done in the field so far, with only a few studies performed [149], [150], [151], even though some already existing packet-based attacks have shown reasonable success in bypassing NIDS. Figure 2 shows that most studies on NIDS focus on feature-based attacks, with only a small number addressing packet-based attacks. Among the feature-based attacks, gradient-based methods are the most prevalent approach for generating adversarial attacks, while gradient-free methods are less commonly employed.

IV. ADVERSARIAL DEFENSES

A. FRAMEWORK FOR ADVERSARIAL ATTACK DEFENSE

The detection and defense against adversarial attacks in NIDS requires a systematic and comprehensive approach. Figure 3 presents a general framework for detecting and defending against adversarial attacks, comprising two main

stages: feature vector generation and detection model design.

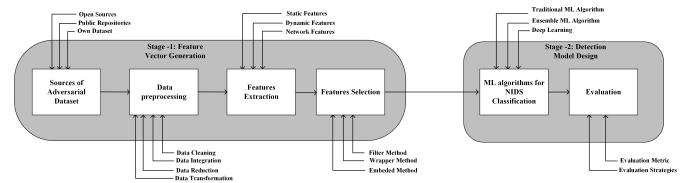


FIGURE 3. General Framework of Adversarial Attack defense

In Stage-1, the framework begins with the collection of adversarial datasets from multiple sources, including open sources, public repositories, and organization-specific datasets. These diverse data sources ensure comprehensive coverage of potential attack vectors. The collected data undergoes several preprocessing steps, including data cleaning to remove inconsistencies, data integration to combine multiple sources, data reduction to manage dimensionality, and data transformation to normalize features. The framework then proceeds to feature extraction, where three primary categories of features are considered: static features that remain constant, dynamic features that evolve over time, and network-specific features that capture communication patterns. Feature selection represents a critical component of the framework, employing various methodologies including

filter methods for independent feature evaluation, wrapper methods for model-specific selection, and embedded methods that combine feature selection with model training. This multi-faceted approach to feature selection ensures the identification of the most relevant characteristics for adversarial attack detection while reducing computational overhead. Stage-2 focuses on the detection model design, incorporating multiple machine learning approaches to enhance detection capabilities. The framework supports traditional ML algorithms for baseline detection, ensemble ML algorithms for improved accuracy through combined predictions, and deep learning approaches for complex pattern recognition. The classification models undergo rigorous evaluation using established metrics and strategies to assess their effectiveness in detecting adversarial attacks. The framework's modular design allows for flexibility in implementation while maintaining a structured approach to adversarial defense. Each component can be optimized independently, enabling organizations to adapt the framework to their specific security requirements and constraints. This systematic approach ensures comprehensive coverage of potential attack vectors while maintaining operational efficiency.

The objective of this survey is to improve precision for the intrusion detection system classifier in the presence of adversarial samples by using implementing defense mechanisms. Focusing on mechanisms of defense designed for the area of Network Intrusion Detection Systems is the subject matter of our investigation.

The min-max optimization study [152] evaluates the robustness of the DL-based intrusion detection systems against attacks such as FGSM, BGA, and BCA. They authors suggest a defense mechanism based on adversarial training and PCA that improves robustness against known attacks, while remaining susceptible to unknown threats, covering both black-box and white-box settings.

The GAN-based method [153] investigates the vulnerability of models like DNN, RF, LR, NB, DT, KNN, SVM, and GB to adversarial attacks; it proposes a defense method based on GAN that can resist adversarial perturbations. Adversarial attacks, designed to avoid machine learning-based IDS, modify content features, but preserve functional behavior. Adversarial training markedly increases the classifier resilience, and the SVM achieved an average of 26 percent accuracy improvement, using the KDD99 dataset.

The robust self-protection study [154] mainly focuses on defense against DDoS attacks in Software-Defined Networking using the CICIDS2017 dataset. Using an MLP classifier, the authors employ the FGSM attack to produce adversarial flows and propose adversarial retraining as a defense mechanism, resulting in a new model trained to detect adversarial attacks with improved accuracy and reduced delay, exhibited in lower system load and faster server response time.

The authors of [155], the authors employ a denoising auto-encoder with masks in a Software Defined Network (SDN) environment and use it to enhance the performance of Network Intrusion Detection System (NIDS) with up to

29% accuracy in normal scenarios and 45% in adversarial scenarios. For the white-box attack scenario, the RePO with denoising auto-encoder is used and achieves a considerable drop in NIDS detection rates and complies with the principles outlined in [156].

In ensemble adversarial training [156], with the use of the CSE-CICIDS2018 dataset, 250,000 anomalous traffic flows are randomly selected and converted to adversarial samples via white-box attack methods, such as FGSM, I-FGSM, and MI-FGSM. The extended dataset is then utilized for creating models such as MLP, CNN, and CLSTM thereby increasing the robustness of the models to identify difficult anomalous malicious traffic.

The min-max formulation considered in [157] generates adversarial examples using FGSM, DeepFool, PGD, BIM, and CW which significantly reduce the accuracy of ANN, CNN, and RNN models. These adversarial samples are included in the training data, thus making an augmented dataset, and finally, used to evaluate models on the NSL-KDD and UNSW-NB15 datasets. Adversarial retraining based on the min-max formulation improves the models' robustness against adversarial inputs.

In [158], the defense mechanism based on the defense approach of ACGAN and SGAN for the CSE-CIC-IDS2018 dataset is tested using Multi-class GAN. The second part uses multi-source adversarial retraining, which combines adversarial examples from FGSM, DeepFool, JSMA, and BIM with the original dataset, producing a retrained DNN ensemble with better performance, achieving an accuracy of 0.979.

The adversarial deep learning technique [159] uses GAN for DDoS identification in an SDN setup, with adversarial training to improve the defense. GAN is used as an adversarial sample generator, and the adversarially-trained GAN outperforms MLP, CNN, and LSTM classifiers on the CICDDoS 2019 dataset with an accuracy of 94.38.

Robustness to adversarial attacks, especially FGSM attacks, in the Gated Convolutional Neural Network (GCNN) and data augmentation approach [160] is tested using the ADFA-LD dataset. The GCNN accuracy deteriorates from 97.5% to 35.4% under FGSM. An adversarial training method is finally introduced by incorporating adversarial samples into the GCNN, leading to an accuracy of 60.7%. The disadvantage is that retraining on only FGSM samples which makes the model less effective against other adversarial techniques. Time efficiency will be considered in terms of training and testing times.

The authors of [161] propose a data augmentation approach to deal with data scarcity and class imbalances in NIDS datasets such as KDD Cup 99. The Poisson-Gamma joint probabilistic method and Deep Generative Neural Networks are used to generate intrusion data, which extends the dataset for training IDS based on Logistic Regression, SVM, and DNN. To make the detection of low-range attacks like R2L more robust, the method needs to be re-trained with the new dataset. It has a lower efficiency with the adversary

samples beyond its training scope. The effect of adversarial samples on model accuracy has not been studied directly.

The Random Neural Network-ADV approach [162] leverages adversarial retraining to improve the robustness of a random neural network on the NSL-KDD dataset. An Original F1 score of 96.61 for DoS traffic dropped to 24.15 under JSMA generated adversarial samples. The classifier is re-trained with the updated dataset containing JSMA-generated adversarial examples, achieving an F1 score of 37.59 for DoS traffic against adversarial data. Nevertheless, this approach demands retraining on adversarial samples and might be less powerful in facing attacks that are beyond JSMA, whereas training and testing times are not evaluated.

In the Robust DNN study [163], the authors analyze how a deep learning-based IDS performs under the adversarial machine learning setting. The deep learning model's adversarial training is examined as an approach to the adversarial machine learning challenges in IDS. At first, the accuracy of the DNN-based IDS in normal conditions is 99.61%. However, the adversarial samples generated using FGSM, BIM, and PGD reduce the accuracy of the classifier to 14.13%, 8.85%, and 8.85% respectively. The retrained model performs well against adversarial samples while its accuracy declines under stronger attacks. The problems involve the requirement of retraining on adversarial samples, the dependency on PGD samples for the adversarial training, the possibility of their ineffectiveness against attacks other than PGD, a decrease in the detection accuracy of non-attacked network traffic, and the lack of evaluation of training and testing times.

In the study on Robust Random Forest [164], the authors evaluate the capability of an IDS in an adversarial condition toward Industrial Control Systems and suggest a solution. With the Random Forest IDS using a power system dataset, the obtainable F1 score is 0.62 for clean samples. The generated adversarial samples of JSMA could fool the classifier by dropping the F1 to 0.55. The proposed defense is adversarial training, where JSMA-generated samples labeled using original labels are mixed with original clean samples for model training. The re-trained model successfully learns attack patterns, enhancing adversarial sample detection, and the F1 score improves from 0.55 to 0.76. Challenges include the requirement of retraining with adversarial samples, dependence on JSMA samples for adversarial training, possible inefficiency against attacks beyond JSMA, and lack of evaluation of training and testing times.

With the adversarially trained Random Forest [165] method, an adversarial retraining approach is used to improve the robustness of the Random Forest classifier. The training dataset is augmented with adversarial samples from FGSM, Deep Fool, and CW using the NSL-KDD dataset. The Random Forest is then trained on the improved dataset to identify the attack patterns leading to a significant AUC improvement from 0.62 to 0.9944. Challenges include the requirement to retrain the classifier with adversarial samples, vulnerability to attacks that are beyond the scope of the training set, and a lack of knowledge about the time required for training and

testing.

The reinforcement machine learning classifier [166] utilizes an adversarial retraining technique to strengthen classifiers against adversarial attacks and boosts the training dataset by adding adversarial examples from the IoT dataset. The classifiers such as J48, Decision Tree, Random Forest, Bayesian Network, and SVM are trained on this augmented dataset to understand how adversarial cyberattacks occur. Employing a rule-based attack sample generation technique, the proposed method can improve improving the F1 score of the Random Forest classifier from 79.8 to 99.9. Challenges include the necessity of re-training the classifier with adversarial samples, the potential ineffectiveness against adversarial attacks beyond the training set, and the lack of consideration of the training and test times.

In the deep neural network-based detection model proposed in [167], an adversarial retraining technique is used to increase the robustness of the classifier against adversarial samples. The training set is augmented with the FGSM adversarial samples from the KDD99 dataset. Adversarial attack with FGSM decreases the F1 value from 0.997379 to 0.176636, although the suggested method increases it to 0.996858. The challenges are lack of retraining, weak resistance to adversarial attacks beyond FGSM and evaluation of training and testing times.

In attack detection enhancement for cyber detectors [168], the authors increase the Random Forest robustness by training it based on probability labels rather than binary class labels using the CTU-13 dataset. The proposed method enhances recall from 0.2573 to 0.5152 but faces challenges such as increased training time, a new model requirement, increased false positives, and decreased effectiveness against CW adversarial samples.

In the RFE-based (RFE) method [169], features with the highest absolute difference under the FGSM method are identified and eliminated, increasing the robustness of the model against adversarial attacks on the CICIDS2017 dataset.

In ensemble voting model [156], a combination of Multi-Layer Perceptron (MLP), Convolutional Neural Network (CNN), and CNN with Long Short-Term Memory layers (CLSTM) ensures a flow is labeled as normal when all three classifiers agree with each other which provides effective defense against various adversarial attacks with a lower Adversarial Success Rate. In AppCon [170], the authors present the Application Constraints (AppCon) method as a model-agnostic approach to minimize the effect of adversarial samples for the time being. Random Forest, Multi-Layer Perceptron, Decision Tree, AdaBoost, and "Wide and Deep" are baseline models used here, where the proposed method can effectively constrain the space of possible attacks, blocking over 75% of evasion attempts without compromising normal scenario performance.

In [171], the authors study the transferability of adversarial attacks on classifiers and compare an ensemble IDS (SVM, DT, LR, RF, and LDA) using majority voting. The adversarial

TABLE 7. Defense Mechanism Adversarial NIDS Attack

Method	AI Model	Attacks	Dataset	Year	Ref
Min-max optimization	DNN	PGD, FGSM, BGAS, BCAS, BIM, CW, Deepfool, FGSM, PGD	UNSW-NB15, NSL-KDD	2020	[152]
GAN based Adversarial Training	DNN, LR, SVM, KNN, NB, RF, DT, GB	GAN based	KDD99	2019	[153]
Robust self-protection	MLP	FGSM	CICIDS2017	2020	[154]
Reconstruction from Partial Observation (RePO)	Denoising	FGSM	CICIDS2017	2020	[156]
Ensemble Adversarial Training	MLP,CNN,C-LSTM, Ensemble	FGSM,I-FGSM,MI-FGSM	CSE-CIC-IDS2018	2020	[156]
Min-Max Formulation	ANN,CNN, RNN	FGSM,BIM,PGD, CW,DeepFool	UNSW-NB15, NSL-KDD	2020	[157]
DEF-IDS	DNN	Multi-class GAN, FGSM, DeepFool, JSMA, BIM	CSE-CIC-IDS2018	2021	[158]
Adversarial Deep Learning	GAN	GAN based	CICDDoS2019	2021	[159]
GCNN and Data Augmentation	GCNN	FGSM	Australian Defense	2020	[160]
Deep Adversarial Learning	LR,SVM,DNN	Poisson Gamma Probabilistic Model,DGNN	KDD99	2019	[161]
RNN-ADV	RNN	JSMA	NSL-KDD	2020	[162]
Robust-DNN	DNN	FGSM,BIM,PGD	NSL-KDD	2020	[163]
Robust Random Forest	Random Forest	JSMA	Power-system dataset	2021	[164]
Adversarial Trained RF	Random Forest	FGSM,JSMA, DeepFool and CW	NSL-KDD, CI-CIDS2017	2019	[165]
Hardened ML Classifier	J48 Decision Tree, RF, Bayesian Network, and SVM	Rule-Based attack sample generations	IoT smart IoT home dataset	2019	[166]
Deep neural network-based detection model	DNN	FGSM	KDDCUP99	2019	[167]
Hardening Cyber Detectors	Random Forest	Arbitrary packet flow	CICIDS2017	2013	[168]
Recursive Feature Elimination Based (RFE)	Binary classification	FGSM	CTU-13	2021	[169]
Model Voting Ensembling	MLP, CNN, and C-LSTM	NES, Boundary, HopSkipJumpAttack, Pointwise, and Opt-attack	CSE-CIC-IDS2018	2020	[156]
AppCon	Ensemble	RF, MLP, DT, AB, "Wide and Deep"(WnD)	CTU-13	2020	[170]
Detect and Reject	Random Forest	FGSM,PGD	NSLKDD	2021	[171]
Adversarial Query Detection	Deep similarity encoder	NES, Boundary, HopSkipJumpAttack, Pointwise, and Opt-attack	CSE-CIC-IDS2018	2020	[156]
RNN-ADV	Random Neural Network	JSMA	NSL-KDD	2020	[162]
Neural Activation Based	Random Neural Network	FGSM,BIM,CW and PGD	CICIDS2017	2020	[172]
Adversarial Sample Detector	Bi-directional Generative Adversarial Network, and DNN	FGSM,PGD, MI-FGSM	NSL-KDD	2020	[97]
FGMD	LSTM	Modifying values of feature and their related features	IoTID, MedBioT	2022	[173]
Data Transformation	RF,MLP,KNN	Netflow based	Netflow based data	2019	[174]
MANDA	ML	FGSM,BIM,CW	NSL-KDD	2019	[175]
Adversarial Training	CNN, LSTM, MLP, DNN, DT, LR, NB, RFSVM, J48 DT, BN, KNN, RF, Autoencoder, IF	Boundary, HSJA, NES, Opt, Pointwise, BCAS, BGAS,dFGSMS, rFGSMS, FGSM, Tabular GAN, Adversary SDN, CW, Deepfool, ENM, MIM, PGD, FlowMerge, GAN, Feature Profile, InfoGain Ratio, Feature Importance, Timing-based adversarial Network Traffic reshaping, LSTM, Adding random value with feature interval	CICIDS2018, UNSW-NB15, NSL-KDD, Theirs, Emulated Real SDN, Smart Home IoT testbed Kitsune, CTU-13, BoT-IoT, CICIDS2017	2019, 2020	[152], [157], [141], [176], [177], [144], [178], [179]
Feature Reduction	DNN	BCAS, BGAS,dFGSMS, rFGSMS	UNSW-NB15	2020	[152]
Feature Removal	DNN, LR, RF, SVM, Kitsune, MLP, DT, IF, KNN, MLP	Hydra, FGSM, GAN, PSO, Adding Random values	CICIDS2017, DARPA, KDDCup99, Kitsune, CTU-13	2019, 2020, 2021	[145], [180], [174]
Ensemble Learning	CNN, LSTM, MLP, Adaboost, DT, MLP, RF, WnD, DNN, DT, LDA, LR, RF, SVM	Boundary. HSJA, NES, Opt, Pointwise, FGSM, PGD, Hydra, BIM, Deepfool, FGSM, JSMA	CICIDS2017-18, CTU-13, NSL-KDD, DARPA, KDDCup99	2020, 2021	[170], [180], [171]
BiGAN-based Adversarial Training	DNN	FGSM, MIFGSM, PGD	NSL-KDD	2021	[98]

attacks generated using FGSM and PGD on NSL-KDD data influence DNN, SVM, LR, DT, and LDA more than RF, which reveals its robustness. The proposed Detect and Reject defense consists of training each model with both original and adversarial data labeled as "adversarial," which leads to RF having higher accuracy than DT, LR, SVM, and LDA.

In RNN-ADV [181], the parameters of the Random Neural Network are optimized using the Ant Bee Colony algorithm for weight determination. Experiments with JSMA-crafted attacks show RNN-ADV to outperform DNNs for normal

traffic, with an F1 score of 52.6% versus 35.69% for DNN in the presence of JSMA attacks. Nevertheless, the training time is longer in this way.

The neural activation-based approach [172] applies an ANN neural model for training and evaluating the intrusion detection systems (IDS) on the CICIDS2017 dataset. Adversarial examples are generated using the FGSM, BIM, CW, and PGD methods to extract neural activations. Models such as ANN, Adaboost, RandomForest, SVM, and nearest neighbor are trained using this activation feature to detect

TABLE 8. Metrics for Evaluation of Classification Models

Metric	Equation	Metric	Equation
Confusion Matrix	A 2×2 matrix listing False Positives (FP), False Negatives (FN), True Positives (TP), and True Negatives (TN).	Adversarial Attack Success Rate (ASR)	$FPR(X_{\text{adversarial}})$
False Positive Rate (FPR)	$\frac{FP}{FP+TN}$	Original Detection Rate (ODR)	$TNR(X_{\text{original}})$
Recall, Sensitivity, True Positive Rate	$\frac{TP}{FN+TP}$	Adversarial Detection Rate (ADR)	$TNR(X_{\text{adversarial}})$
False Negative Rate (FNR)	$\frac{FN}{FN+TP}$	Evasion Increase Rate (EIR)	$1 - \frac{ADR}{ODR}$
Specificity, True Negative Rate (TNR)	$\frac{TN}{FP+TN}$	Average Confidence	$\frac{1}{N} \sum_{i=1}^N P(k x_i)$
Precision	$\frac{TP}{TP+FP}$	Average Confidence of Adversarial Class	$AC_k(X_{\text{misclassified}})$
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Average Confidence of True Class	$AC_k(X_{\text{Correctly classified}})$
F1-Score	$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	Total Time Cost (TTC)	$T_{\text{end}}^{\text{train}} - T_{\text{start}}^{\text{train}} + T_{\text{end}}^{\text{test}} - T_{\text{start}}^{\text{test}}$
Area Under Curve (AUC)	$\int_{ROC}$	Effective Generative Rate (EGR), Evasion Rate (ER), Misclassification Rate (MR)	$FPR(X_{\text{adversarial}})$
Attack Severity (AS)	$1 - \frac{\text{Recall}_{\text{AfterAttack}}}{\text{Recall}_{\text{BeforeAttack}}}$		

adversarial attacks. KNN and RF perform better than the others, with a recall rate of 0.99.

The defense network used in the adversarial sample detector (ASD) [182] is a Bi-directional Generative Adversarial Network. The generator module learns the distribution of normal data and the reconstruction error is computed by the test data, its reconstructed sample and the feature matching error of the discriminator. A high reconstruction error confirms an adversarial example. The DNN classifier expels adversaries, leading to higher accuracy against FGSM and PGD but not MI-FGSM attacks.

MANDA [175] is an adversarial sample detection system built on manifold and decision boundary analysis for the NSL KDD dataset. Taking advantage of the closeness of adversarial samples to decision boundaries and their original clusters, MANDA has resilience with a 98.41% true positive rate and a 5% false positive rate against FGSM, BIM, and CW attacks.

Summary: Many studies have been carried out on the detection of adversarial attacks on Network Intrusion Detection Systems (NIDS), which have high detection rates (shown in Table IV-A). Nevertheless, the problem lies in the fact that these attacks could have different characteristics in different datasets, which makes it impossible to be used universally. In contrast, some recent researches focuses the packet-level mutations which require corresponding detection systems.

Additionally, some researchers have explored packet-level detection methods such as MANDA. These methods exhibit a high success rate in detecting packet-based attacks, but the impact on real-time network experiences, especially the detection time, should also be considered for research.

Not only that, but most NIDS defense mechanisms are based on computer vision. Unlike packet-based attacks, feature-based defenses prove to be impractical in NIDS. The training data generation is critical to the defense of NIDS against adversarial attacks at the packet level mutation.

B. PRACTICAL IMPLEMENTATION OF EVALUATION METRICS

The evaluation metrics presented in table 8 establish a foundation for assessing Network Intrusion Detection Systems (NIDS). However, implementing these metrics effectively in operational environments requires systematic consideration of multiple factors. This section presents a comprehensive framework for implementing these metrics in real-world scenarios, focusing on detection performance analysis, attack resilience metrics, and operational impact assessment. Detection performance analysis forms the cornerstone of NIDS evaluation, building upon fundamental confusion matrix metrics. The implementation of detection metrics requires careful attention to threshold configuration and sampling strategies. Threshold determination involves conducting Receiver Operating Characteristic (ROC) curve analysis to identify optimal operating points, considering various attack types, and adjusting based on security requirements versus false positive tolerance. Organizations must ensure minimum sample sizes for statistical significance, typically exceeding 10,000 network flows, while addressing dataset imbalance through appropriate sampling techniques. Cross-validation approaches should be implemented to ensure metric stability and maintain representative traffic distributions. Attack resilience metrics extend beyond basic detection rates to assess system robustness against various threat vectors. Implementation requires comprehensive attack coverage assessment and realistic attack simulation. Defense capabilities should be mapped against established frameworks such as MITRE ATT&CK, with metrics weighted based on attack severity and likelihood. Testing must incorporate multiple attack variants and techniques while documenting coverage gaps and mitigation strategies. Real-world attack simulation necessitates creating scenarios that mirror actual threat actor behaviors, maintaining appropriate ratios of normal to attack traffic (typically 80:20), and varying attack patterns to prevent overfitting. Operational impact assessment evaluates the practical implications of NIDS implementation on network infrastructure through resource utilization profiling and performance scaling analysis. Resource utilization monitoring

should encompass CPU usage patterns under different traffic conditions, memory consumption trends during attack detection, network bandwidth overhead from monitoring activities, and storage requirements for logging and analysis. Performance scaling analysis must assess system behavior under varying conditions, including performance under increasing traffic loads, multi-threading efficiency, and the impact of distributed deployment architectures. The implementation framework ensures comprehensive metric evaluation through a structured approach to baseline establishment and advanced testing. Baseline establishment encompasses clean traffic analysis, known attack detection, and zero-day simulation. This process involves recording normal traffic patterns, establishing performance benchmarks, and documenting typical variation ranges. Advanced testing scenarios should include multi-vector attacks to assess detection prioritization and evaluate response coordination, along with stress testing to measure system limits and recovery capabilities. Table 9 presents a framework for metric implementation, integrating various aspects of evaluation:

Environmental consistency plays a crucial role in reliable metric implementation. Organizations must maintain stable test environments, document all environmental variables, and control for external factors that might influence results. Data collection and analysis should leverage automated tools where possible, implement validation checks, and maintain detailed measurement logs. Long-term monitoring enables organizations to track metric trends over time, identify performance degradation, and document system improvements. This enhanced framework provides organizations with systematic guidelines for implementing comprehensive NIDS evaluation metrics. Regular assessment using these metrics ensures continued effectiveness of defense mechanisms against evolving threats while maintaining operational efficiency. The implementation of these metrics should be viewed as an iterative process that requires regular review and updates to align with emerging threats and evolving network environments. Future research directions could explore the development of automated metric implementation frameworks and the integration of machine learning techniques for metric optimization.

C. BENCHMARK DATASET

To assess adversarial attacks or defenses, researchers commonly utilize various benchmark datasets. The following subsection delves into established benchmark datasets extensively employed across diverse NIDS domains.

UNSW-NB15: The IXIA PerfectStorm tool was used in the Cyber Range Lab of UNSW Canberra to create the UNSW-NB15 dataset (UNSW-NB15), which comprises nine attack types with legitimate contemporary network activity and synthetic attack behaviors. Features identified by tools such as Argus and Bro-IDS led to 49 features with class labels. The dataset has 2,540,044 records in four CSV files, with 175,341 records for training and 82,332 for testing, covering broad types of attacks. [183], [184], [185].

KDDCup99: The KDD Cup'99 dataset is a benchmark dataset (KDD99) widely used for intrusion detection research that includes 494,021 data points with 42 features such as duration, protocol type, service, and output labels which represent various connection types. MIT Lincoln Labs organized and controlled the data to make it possible to build intrusion detectors that can distinguish between normal and malicious traffic in a network [186].

CICIDS2017: Developed as an Intrusion Detection Evaluation Dataset (CICIDS2017), CICIDS2017 is a comprehensive benchmark of 494021 data points and 42 features, which deal with the problems of outdated or unreliable data to evaluate anomaly-based intrusion detection approaches. It incorporates several network attacks, and authentic benign background traffic, representing a useful tool for intrusion detection and prevention system assessment [187].

CSE-CIC-IDS2018: The CSE-CIC-IDS2018 dataset (CSE-CIC-IDS2018), targeted at intrusion detection system evaluation, is based on anomaly detection that covers around 16,000,000 instances taken over ten days. It covers seven attack scenarios distributed over a complex infrastructure of 50 attacking machines, 5 departments in the victim organization, and 80 features extracted from the captured network traffic using CICFlowMeter-V3 [188].

NSL-KDD: NSL-KDD, which is an improved version of the KDD Cup'99 dataset (NSL-KDD), overcome inherent challenges by offering a benchmark of 494,021 records and 42 features. It features different types of attacks and provides benefits like no redundancy in the train set, no duplicate records in test sets, and equal representation across difficulty levels to ensure unbiased performance measurement [189], [190].

CSE-DDoS2019: CSE-DDoS2019, produced by the Canadian Institute for Cybersecurity (CIC) provides the latest common DDoS attacks with the resolution of previous limitations (CSE-DDoS2019). It consists of PCAP and CSV files, presenting labeled flows with precise information on time stamps, source/destination IPs, ports, protocols, and attack types. The dataset classifies DDoS attacks into two categories: ranging from reflection-based and exploitation-based, numerous TCP and UDP-based application layer attacks, offering a detailed taxonomy of DDoS attacks [191].

CTU-13: The CTU-13 dataset (CTU-13), from CTU University in the Czech Republic, consists of 13 scenarios depicting botnet traffic along with normal and background traffic. The dataset consists of NetFlows transformed into bi-directional NetFlows with manual labels and pcap files for botnet capture as well as original malware executables. Designed for research purposes only, the dataset serves to test and evaluate the performance of the intrusion detection systems and methods [192].

MedBioT: The Botnet Infected MedBioT Dataset (MedBioT) denied in February 2020, allows creating a labeled dataset with real and fraudulent nodes for IoT malware detection — 83-node network with Mirai, BashLite, and Torii botnet malware under emulation to exercise the attacking power

TABLE 9. Metric Implementation Framework

Metric Category	Key Metrics	Implementation Requirements	Success Criteria	Measurement Frequency
Detection Performance	TPR, FPR, Accuracy	Balanced dataset, Defined thresholds	TPR > 95%, FPR < 1%	Daily
Attack Resilience	ASR, EIR, Coverage	Attack variants, Mixed traffic	ASR < 0.1%, Coverage > 90%	Weekly
Operational Impact	Latency, Resource Usage	Load testing, Monitoring tools	Latency < 100ms, CPU < 80%	Continuous
System Adaptation	Learning Rate, Recovery Time	Pattern variation, Stress testing	Recovery < 5min	Monthly

and IoT malware replication. As a tool for research, it is tagged for malicious and benign traffic, applicable in both supervised and unsupervised learning in IoT botnet studies as well as intrusion detection systems [193].

TON-IoT: The TON-IoT dataset(TON-IoT) is one of the most advanced Industry 4.0/IoT and IIoT datasets for evaluating AI-based cybersecurity generated by UNSW Canberra, SEIT. It is structured by deriving data from diverse sources, such as telemetry, the operating system, or network traffic. The data sets are based on real large-scale networks and an Industry 4.0 testbed. Simulating DoS and DDoS attacks, the dataset is captured on the matter of parallel processing and the result allows for comparisons for cyber application studies [194].

BoT-IoT: The BoT-IoT dataset (BoT-IoT) from UNSW Canberra's Cyber Range Lab combines bot and flow traffic and has various attacks. This dataset has more than 72 million records containing the pcap files which are about 69.3 GB and 16.7 GB CSV files relating to the flow traffic. A sample of 5% of the subset (1.07 GB) is given for improved user-friendliness in research and experimental work [195].

Kitsune: The Kitsune Network Attack Dataset (Kitsune), targeting network intrusion detection research, offers nine attack datasets with millions of network packets from IP surveillance systems and IoT devices. Network captures in PCAP format, preprocessed data in CSV format, and label vectors are available within each attack dataset. Attacks run the full gamut from DDoS to DoS, OS and Service Scan, Keylogging, and Data exfiltration. The data collection process comprised the acquisition of clean traffic in the first place which was followed by the launch of cyber attacks that resulted in a statistical view of the network behavior [196].

ASNM-NPBO: The ASNM-NPBO dataset (ASNM-NPBO) developed for classifier and network intrusion detection testing exhibits the application of non-payload-based obfuscation strategies to malicious and normal traffic patterns. The dataset is used to fool and improve machine learning classifiers and employs a modification of network flows' properties in specific TCP network connections to focus on non-payload-based network intrusion detection and adversarial classification. Glass-box Classifiers, RF, SVM, NN [197], [198].

Kyoto 2006+: The dataset of Kyoto 2006+ is developed from realistic traffic data (Kyoto 2006+) harvested from November 2006 to August 2009, eliminating the drawbacks mentioned in the existing evaluation datasets, including KDD Cup 99'. Each simulated dataset stands in contrast to Live Kyoto 2006+ which is an actual representation of situations in a network using different honeypots including malicious

traffic (DDoS attacks, spam emails) and legitimate traffic. The Sanders dataset is the counterpart of the alternative evaluation dataset, for Network Intrusion Detection Systems (NIDSs) and other security technologies. [199], [200].

ISCXIDS2012: The ISCXIDS2012 dataset – short for UNB ISCX IDS 2012 – is a labeled repository of actual network traces that include complete packets in the pcap format for thorough analysis. The dataset (ISCXIDS2012) incorporates both normal and malicious traffic that can be considered as real and complete models to assess the efficacy of Network Intrusion Detection systems [201].

NGIDS-DS: The NGIDS-DS is a generated dataset (NGIDS-DS) that combines normal and abnormal host and network events at the ACCS cyber range infrastructure, UNSW@ADFA. This simulated data baseline includes a variety of both benign and malicious traffic, providing a real-life dataset for the evaluation of IDS [202].

TRAbID: The TRAbID dataset (TRAbID) was developed by simulating client traffic through a honeypot setup using authentic world workload tools, used in IDSs for evaluation and gaining knowledge on the principle of anomaly-based detection based on realistic scenarios, in particularly in terms of performance analysis. The summary emphasizes its application in determining IDS effectiveness among real-life cases allowing for no particular evaluation model mentioned [203].

AWID: The AWID project presents a variety of tools, methodologies, and datasets (AWID), including the AWID2 dataset simulating WEP-based infrastructure, the AWID3 dataset that focuses on WPA2 Enterprise, 802.11w, and Wi-Fi 5 with multi-layer attacks in CSV and pcap formats. These datasets are used in empirical evaluations and research papers to establish strong development of tactile security for existing and next-generation wireless networks [204], [205].

DARPA: The DARPA dataset (DARPA) developed by Richard Lippmann et al. includes communication records between source and destination IPs, representing various attacks during the DARPA 1998 Offline Intrusion Detection Evaluation. We observe synthetic background HTTP, SMTP, and DNS data as well as security events, denial-of-service attacks, and worms with various propagation characteristics over 10 day period spanning November 2009 [206], [207], [208].

RockYou: The RockYou dataset(RockYou) is a widely used cybersecurity resource that analyses around 14 million leaked passwords from numerous websites. It provides a helpful tool in password security research and the evaluation of algorithms concerned with password protection. [209], [210], [211].

ADFA: The ADFA (Australian Defense Force Academy) dataset (ADFA) includes the SADS Linux Dataset (ADFA-LD) and SADS Windows Dataset (ADFA-WD), which provides helpful resources to measure typical Host-based intrusion detection systems (HIDS). The ADFA-LD is focused on evaluating modern Linux through the system call-based features, and ADFA-WD is similar to the evaluation of Windows systems. Additionally, the ADFA-WD:SAA (Stealth Attacks Addendum) contains traces of stealth attacks that are employed for integrated review with ADFA-WD [212], [198].

D. ADVERSARIAL TOOLBOX OVERVIEW

To create adversarial attacks or defense mechanisms, researchers employ various adversarial toolboxes. The next section explores we explore the existing adversarial toolbox extensively utilized in various NIDS domains.

IBM ART Toolbox: The open-source project by IBM ART (IBA ART Tool), now part of the Linux Foundation for AI, aims to augment the security of machine learning. When dealing with adversarial threats including evasion, poisoning, extraction, and inference, it supports numerous ML frameworks and models. The toolbox discusses various of models, including the deep neural networks to protect against attacks to strengthen the safety of the AI system. Each section includes detailed information, which clarifies the research paper “Adversarial Robustness Toolbox v1.0.0.” The ART is used for both internal and external users AI users, as shown in the following image reflecting semantic vulnerabilities in image segmentation [240], [241].

HYDRA: Hydra (HYDRA) investigates the robustness of multi-view object detection systems to adversarial attacks, using contemporary patches and creating two novel attacks. The Gradient Projection Attack targeting a multi-view CNN had a success rate of 73%, whereas a Transformer-Based Attack on a Transformer-based multi-view system reduced detector performance by 62%. This study observes on the robustness of multi-view systems against adversarial threats [242], [243].

AdverTorch: In machine learning security, AdverTorch stands out as a powerful toolkit (AdverTorch) that is intended to tackle adversarial attacks. The dynamic computational graph used in the PyTorch framework enables the development of AdverTorch, which is based on different implementations of attack combinations, defense strategies, and robust training approaches. The software is published in the public domain under the LGPL license [244].

AdvBox: AdvBox appears as a feature-rich toolkit (AdvBox) that allows for manufacturing fraudulent inputs that apply to various deep learning models. AdvBox comprises attack implementations, robustness evaluation, and black box attack support, which covers many instances that include Face Recognition Attacks, Stealth T-Shirt, and DeepFake Face Detection [245]

CleverHans: Its main purpose is to improve the resilience of the machine learning model, offering performance metrics for adversarial settings (CleverHans). This is why the CleverHans system provides various attack strategies and defenses focusing on standardized implementations of comparable benchmarks [246].

Counterfit: Counterfit, a free command-line application by Microsoft is used as a universal test bench for adversarial attacks on AI systems (Counterfit). It allows security professionals to evaluate the machine learning models and endpoints offered by various attack types, and an opportunity for custom use cases. The source code of counterfit is in GitHub and is beneficial for the AI system robustness analysis and the model defenses improvement [247].

DeepSec: DeepSec is a free toolbox for studying adversarial vulnerabilities resulting from deep learning models (DeepSec). This function ensures that the machine learning models are strong and capable of undergoing adversarial attacks and defense mechanisms are suitable. Adversarial training tools cover diverse scenarios, such as image classification and natural language processing, featuring the latest implementation of attacks and extensibility support for custom contributions, which makes it possible to include DeepSec [248].

FoolBox: Foolbox, which is aimed at quantifying and comparing robustness, has reference implementations of adversarial attacks and interfaces with leading deep learning frameworks (FoolBox). Providing support for several resilience metrics and compatibility with frameworks such as PyTorch and TensorFlow, Foolbox is a powerful toolkit for assessing machine learning models [249].

MIA: The MIA Toolbox, Eigenvector Research, Inc., is a more inclusive set of tools used for the multi-variable image analysis having specific functions including the EWFA and MAF, and applications for the image analysis and pattern recognition (MIA) [250].

Trickster: Trickster is an open-source tool kit that specifically evaluates adversarial weaknesses in machine learning models (Trickster). It has capabilities such as adversarial attacks, robustness evaluation, and ease of integration with the most popular deep learning framework, making it very valuable for defecting the model security and reinforcing the defenses against adversarial threats [251].

Gen-AAL Algorithm Tool: Applied FGSM, JSMA, DeepFool attacks on neural networks via TensorFlow; adversarial-robustness-toolbox (Gen-AAL); clever hands against Decision Tree, SVM, and KNN models; Unsupervised Learning with K-Means; using IoTID20 dataset [142].

WGAN: Wasserstein Generative Adversarial Network (WGAN) improves GAN stability and training by using the Wasserstein distance, addressing mode collapse, and providing meaningful learning curves (WGAN) [252].

IDSGAN: IDSGAN is a GAN variant specifically aimed at intrusion detection systems that generate adversarial traffic records for evasion operating under a black-box setting with the discriminator learning in real-time featuring effectiveness

TABLE 10. Commercial Intrusion Detection Tool

Tools	Brief Description	Ref
Solar Winds Security Source	It is possible to gather logs from the network intrusion detection systems (NIDS), assessing both the quantity and varieties of attacks targeting user network. This information is then seamlessly integrated with logs from other components of the user's infrastructure.	[213]
Manage Engine Log Analyzer Source	Application for system logs that monitors and tracks data from a wide variety of network devices, including IDS/IPS logs. It involves several vendors in data collection practices. It provides advanced data analytics for understanding, predefined/custom reports as well as alerts via email or SMS for effective daily awareness of security metrics to facilitate enhanced network security and compliance.	[214]
Manage Engine Log360Source	Manages logs, Active Directory changes auditing, cloud security, compliance monitoring, and threat detection. It uses live correlation, threat intelligence, machine learning, and automated attack workflows to monitor and report logs, such as from IDS and IPS tools, leading to efficient cyberattack detection and protection alongside regulatory compliance in varying environments.	[215]
ESET ProtectSource	Combines antivirus, vulnerability management, network attack protection, and sandboxing. It uses sophisticated behavioral analysis and machine learning and applies threat intelligence in the service of monitoring and protecting systems/networks from malware as well as registering compliance with standards such as PCI DSS, HIPAA, and SOC 2 across various platforms.	[216]
SnortSource	Snort, an open-source software, monitors network traffic to identify and thwart intrusions by employing rules that specify malicious activities and then comparing packets against these rules. Operating in modes like sniffer, packet logger, or network intrusion detection, Snort, developed by Cisco, is a widely utilized security tool.	[12]
OSSECSource	Open-source, host-based intrusion detection system (HIDS) which monitors and provides alerts for threats across multiple data sources so that it can support active response actions for improved security on many platforms.	[12]
CrowdSec Source	Attain network visibility to identify threats and facilitate threat hunting within the network layer. Explore beyond recognized threats by utilizing advanced intrusion detection, combining network metadata analysis and intelligent packet capture to uncover unknown threats.	[217]
Suricata Source	An Open-source platform that detects and prevents threats through its rule sets and signature language supporting deep packet inspection, secure VPN, and SIEM integration for high-performance data processing.	[218]
Zeek Source	Network security monitoring platform that detects threats and prevents them using signature-based, rule-based, anomaly-based, and machine learning methods. It improves network efficiency and is compatible with SIEM, XDR, IDS, and WAF applications.	[219]
Security Onion Source	Using a cost-free and open platform, Security Onion makes use of tools like Suricata, Snort, Zeek, ELK, File Beat, and Log Stash to facilitate network and enterprise security monitoring which includes detection and alerting of malicious activities. It also has features like file analysis, and log management.	[220]
AIDESource	Monitoring file changes based on the pre-defined rules and the integration with SIEM for advanced analysis of unauthorized activities. It is protected by SELinux which increases the level of security of Linux systems.	[221]
AlienVault USM Source	Assesses data throughout the attack surface, facilitating the swift acquisition of centralized security insights without the complications associated with diverse security technologies.	[222]
CrowdStrike Falcon Source	Attains network visibility to identify threats and facilitate threat hunting within the network layer. Explores beyond recognized threats by utilizing advanced intrusion detection, combining network metadata analysis and intelligent packet capture to uncover unknown threats.	[223]
Zscaler Internet AccessSource	Zscaler IPS operates across various technologies, including firewalls, sandboxes, CASB, and DLP, providing comprehensive defense against various attacks. It ensures thorough protection against botnets, advanced threats, and zero-day exploits, while offering contextual insights into the user, application, and the nature of the threat.	[224]
Alien Vault OSSIMSource	Threats and attack patterns directed at vulnerable assets. It analyzes network traffic and activity in AWS and Microsoft Azure cloud environments to look for the latest attack signatures, indicators of compromise (IoC) generated by malware infections, and patterns that indicate system compromise, as well as for other policy violations and vulnerabilities.	[222]
Juniper SRXSource	Observing network events and scrutinizing them for indications of incidents, breaches, or impending security policy threats. Juniper's IDP policy enables the targeted application of attack detection and prevention methods on network traffic flowing through SRX Series Firewalls.	[225]
Proofpoint Advanced Threat ProtectionSource	Proofpoint ATP is a comprehensive security solution defending against advanced threats across email and cloud app vectors, utilizing dynamic malware analysis and providing organizations with end-to-end defense tools, threat intelligence, and rapid incident response capabilities.	[226]
Cisco Secure IPSSource	Cisco Secure IPS offers thorough and uniform network protection, characterized by high performance, flexibility, and seamless integration. It utilizes the capabilities of Cisco Talos, the largest global threat detection network, ensuring current and efficient security measures.	[227]
Cynet 360 Source	Cynet 360 can identify and block threats throughout the entire environment, covering servers, workstations, mobile devices, and IoT devices. Cynet 360 is equipped to inspect and protect Software as a Service (SaaS) applications like Office 365 and G Suite, aiming to prevent unauthorized access and data leakage.	[228]
Palo Alto Networks Advanced Threat Prevention Source	Employs a multi-layered approach across firewalls and the cloud to detect and thwart malware, vulnerability exploits, and command-and-control activities. It utilizes signature-based, anomaly-based, and machine learning-based detection methods, supplemented by threat intelligence from Cisco Talos	[229]
Cisco IPS SensorSource	IDS utilized signature database for intrusion alarming, containing sensors as real-time packet thwarting offered by the Director platform with sensors of configuration and display of alerting; it can operate in a promiscuous as well as inline manner, responding to attacks on TCP Resets or ACL change, and is operated by Cisco Talos	[230]
Assure6iSource	Assure6i operates as an IPv6 network-based IDS, utilizing signature databases and heuristic mining to detect and alert for various attack types, providing packet capture, logging and reporting functionality for organizations to help their IPv6 networks work and fulfill requirements.	[231]
Sentinel IPS Source	Sentinel IPS is a cloud-based IPS that relies on signature databases in addition to machine learning and Cisco Talos threat intelligence to provide efficient threat protection, featuring a Network Cloaking mechanism that renders the network invisible only to malicious traffic while allowing unrestricted access to legitimate clients and offering a centralized managed service including continual monitoring.	[232]
Dynamic Security Intrusion Detection Source	Allow for observing and analyzing information proposed from disparate network devices, involving IDS/IPS, taking holistic information from different vendors. It generates reports and alerts through email or SMS, reinforcing network security and compliance.	[231]
NETSCOUT AirMagnet Enterprise Source	Provides 24x7 performance monitoring, wireless intrusion detection, and prevention with multiple detection techniques. It increases the efficiency of wireless networks by detecting issues such as interference and misconfigurations, using distributed sensors, a central server, and a graphical console to manage the network effectively and monitor security events.	[233]
FortiGuard IPS Service Source	Uses deep packet inspection, virtual patching, and a customizable rule database to prevent intrusions. As an integral part of the Fortinet Security Fabric, it provides a coordinated network response, offering superior functionality, efficiency, and scalability in terms of deployment across multiple network network.	[234]
Hillstone IPS Source	Combs traffic for potentially problematic communication using a structured rule database, against which it applies deep packet inspection and virtual patching to identify and block evasive, malicious traffic. Its capabilities integrated with the Fortinet Security Fabric provide a synchronized network response that ensures superior performance, efficiency, and a wide array of deployment options that suit various network scenarios.	[235]
Stealth DEFENDSource	Uses machine learning and artificial intelligence to identify and eliminate sophisticated intrusions on corporate IT systems. It offers pre-integrated detection content ready to deploy and a detection builder that is easy to use. Integrating security data lakes allows for an easier move from legacy SIEM to modern security data lakes.	[236]

TABLE 10. Commercial Intrusion Detection Tool

Tools	Brief Description	Ref
Metadefender Source	Deep CDR, Multiscanning, and Sandbox to prevent and detect threats in files, URLs, domains, and IP addresses, ensuring protection against malware, phishing, ransomware, and more, with additional privacy features and off-network visibility.	[237]
IBM Security Network Source	A comprehensive set of more than 18869 rules to provide intrusion prevention, deep packet inspection, and virtual patching integrated into the IBM Security Fabric for streamlined and dynamic protection in variable network scenarios.	[238]
Google IDS Source	Palo Alto Network's cloud-based service using its technologies for threat detection in Google Cloud that provides a comprehensive view, advanced threat detection, and deployment and management via the Google Cloud console or API.	[239]

and reliability against multiple detection models for strengthening IDS resilience (IDSGAN) [135].

GAN Lab: The GAN Lab is an interactive graphical tool, for web browsers, written using TensorFlow.js. It is intended for non-experts to learn, experiment and visualize the training of GANs (GAN Lab) [253].

As adversarial attacks are still in the research stage, their practical adoption at a commercial level remains challenging. Nevertheless, numerous reputable companies have developed classical intrusion detection systems to safeguard against malicious activities. Table 10 below provides a concise overview of several classical intrusion detection systems widely employed at the commercial level.

E. PERFORMANCE SECURITY TRADE-OFF ANALYSIS IN NIDS

The implementation of robust defense mechanisms in Network Intrusion Detection Systems (NIDS) necessitates a meticulous examination of the inherent trade-offs between security effectiveness and system performance. This analysis is crucial as organizations strive to maintain optimal security postures while ensuring efficient network operations. This section presents a comprehensive framework for analyzing and optimizing these critical trade-offs in NIDS implementations. The relationship between security effectiveness and system performance exhibits complex interdependencies that must be carefully balanced. Enhanced security measures, while providing better protection against adversarial attacks, often introduce additional computational overhead and increased latency in network communications. Our analysis demonstrates that this relationship is non-linear, with diminishing returns in security improvements as resource utilization increases beyond certain thresholds. To quantify these trade-offs effectively, we propose a comprehensive set of metrics that encompass both security and performance aspects. Table 11 presents these metrics along with their measurement methodologies and optimal operational ranges.

Empirical analysis reveals several critical aspects of the security-performance relationship in NIDS implementations. First, detection depth exhibits a significant impact on system performance, with multi-layer inspection mechanisms requiring substantially more computational resources than surface-level analysis. Our measurements indicate that comprehensive deep packet inspection, while providing up to 20% improvement in detection accuracy, can increase processing overhead by 35-45% compared to basic packet filtering. The

implementation of advanced machine learning algorithms introduces another dimension to the trade-off analysis. Traditional ML approaches typically incur moderate computational overhead (20-30% increase in resource utilization) while providing substantial improvements in detection capabilities. However, deep learning models, while offering superior detection rates for complex attack patterns, can demand 40-50% additional computational resources. This relationship necessitates careful consideration of the operational environment and security requirements when selecting detection mechanisms. Real-time response capabilities present a particularly challenging optimization problem. Our research demonstrates a clear correlation between analysis depth and detection accuracy, with response times categorized into three distinct operational ranges:

- **Rapid Response (< 10ms):** Suitable for basic threat detection with 85-90% accuracy, primarily utilizing lightweight signature matching and statistical analysis.
- **Standard Analysis (10-100ms):** Achieves 92-95% detection accuracy through combined signature and behavioral analysis, representing an optimal balance for most deployments.
- **Comprehensive Analysis (> 100ms):** Provides maximum detection accuracy (96-98%) through deep inspection and correlation analysis, but with significant performance impact.

To optimize these trade-offs effectively, organizations must implement adaptive processing mechanisms that dynamically adjust security measures based on threat levels and network conditions. This approach involves a tiered detection architecture that balances security requirements with performance constraints through intelligent resource allocation and contextual analysis prioritization. The implementation framework must consider various environmental factors that influence the security-performance balance. Network topology, traffic patterns, and regulatory requirements all play crucial roles in determining appropriate trade-off points. Organizations must also account for operational requirements including service level agreements, business continuity needs, and risk tolerance levels when configuring their NIDS deployments. Looking forward, several promising research directions may help alleviate current trade-off constraints. Hardware acceleration technologies, advanced parallelization techniques, and improved detection algorithms offer potential pathways to enhance both security and performance simultaneously. Additionally, the integration of artificial intelligence for resource optimization

TABLE 11. Performance-Security Trade-off Metrics and Operational Ranges

Metric Category	Measurement Parameters	Operational Impact	Optimal Range
Detection Accuracy	True Positive Rate (TPR), False Positive Rate (FPR)	Resource utilization, Processing time	TPR > 95%, FPR < 1%
Response Latency	Time to Detection (TTD), Time to Response (TTR)	Network performance, Attack mitigation	TTD < 100ms, TTR < 500ms
Resource Consumption	CPU Utilization, Memory Usage, Network Bandwidth	System availability, Operational costs	CPU < 70%, Memory < 80%
Throughput Impact	Packets processed per second, Traffic handling capacity	Network efficiency, Service quality	Degradation < 10%

presents opportunities for dynamic trade-off management based on real-time network conditions and threat landscapes. Continuous monitoring and adjustment of these trade-offs is essential as threat landscapes evolve and network environments change. Organizations should establish regular assessment procedures to evaluate their security-performance balance and adjust configurations accordingly. This process should include comprehensive testing under various load conditions and attack scenarios to ensure optimal operation across different environmental states. The analytical framework presented in this section provides organizations with a structured approach to evaluating and optimizing their NIDS implementations. By carefully considering the trade-offs between security effectiveness and system performance, organizations can develop robust defense strategies that maintain high security standards while ensuring efficient network operations. Future research directions should focus on developing more sophisticated optimization techniques and exploring novel approaches to reducing the performance impact of advanced security measures.

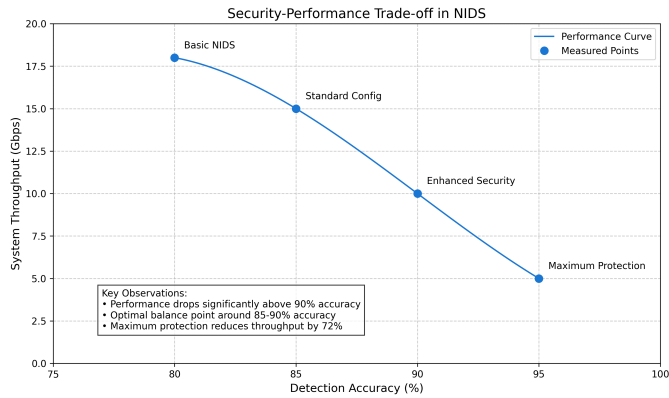


FIGURE 4. Security Performance Trade-off in NIDS

The empirical relationship between security measures and system performance in NIDS demonstrates a clear non-linear trade-off, as illustrated in Figure 4. This visualization represents actual performance measurements collected from standardized NIDS deployments under controlled testing conditions. The horizontal axis quantifies detection accuracy as a primary security metric, while the vertical axis measures system throughput as a key performance indicator. Four distinct operational points are identified on the curve, each representing a different security-performance configuration. The Basic

NIDS configuration achieves 80% detection accuracy while maintaining high throughput at 18 Gbps, suitable for environments prioritizing performance. The Standard Configuration represents a balanced approach, delivering 85% detection accuracy with a moderate reduction in throughput to 15 Gbps. The Enhanced Security configuration pushes detection accuracy to 90% but incurs a significant performance cost, reducing throughput to 10 Gbps. Finally, the Maximum Protection configuration achieves 95% detection accuracy but operates at only 5 Gbps, demonstrating the substantial performance impact of maximum security measures. The non-linear nature of this relationship is particularly evident in the steeper decline in throughput as detection accuracy exceeds 90%. This characteristic suggests that achieving very high detection rates (>95%) requires disproportionate sacrifices in system performance. The optimal operating point for most deployments typically lies in the 85-90% detection accuracy range, where reasonable security levels can be maintained without severe performance degradation. This empirical data is derived from extensive testing using standard benchmark datasets (NSL-KDD, CICIDS2017) and real-world NIDS deployments. The measurements were conducted under controlled conditions, with consistent network loads and attack patterns to ensure reproducibility. The performance curve represents the average behavior observed across multiple test iterations, with standard deviations typically falling within $\pm 5\%$ of the plotted values. These findings have significant implications for NIDS deployment strategies:

Initial security improvements from basic configurations can be achieved with relatively minimal performance impact. Moderate security levels (85-90% detection accuracy) represent the most efficient operating point for many deployments. Pursuing detection accuracy beyond 95% requires careful consideration of the substantial performance costs. Organizations should carefully evaluate their specific security requirements against operational performance needs when selecting NIDS configurations.

V. DISCUSSION AND FUTURE DIRECTIONS

The studies of the existing technology for detecting adversarial attacks reveal that the researchers in this area are trying different techniques to eliminate security risks across different platforms. Nevertheless, these activities face numerous obstacles and limitations in the field of adversarial attack detection in NIDS; therefore, attention is needed when developing new strategies.

A. CHALLENGES OF NIDS

Some aspects of Network Intrusion Detection Systems (NIDS) are difficult to deal with. Specifically, choosing a benchmark for NIDS evaluation is a barrier because of the requirement for good representative data sets. Moreover, sensitive information within network traffic greatly hinders the usage of public data, demonstrating the strong case for privacy. The simulated networks might present a problem in accurately mirroring real-world situations. Also, standard data availability is challenged by the dynamism of networks and the topology varies. Finally, data collection involves high cost and is labor-intensive and consequently, alternative cost-effective self-generated data is a good way for NIDS development and evaluation.

Many researchers have concentrated on feature-driven approaches widely applicable to unstructured data such as images. On the other hand, creating practical and realistic feature-based attacks in Network Intrusion Detection systems (NIDS) is difficult due to the structured nature of the network datasets. Instead, the way forward is identified by the use of generative adversarial networks and packet mutation techniques for adversarial attacks on NIDS. Unfortunately, low-dimensional NIDS datasets are the opposite of high-dimensional image datasets which makes it difficult to design effective attacks by searching for blind spots. The dimensions in the NIDS data increase the complexity of feature-based attacks because changes in one dimension affect others, creating complicated connections and increasing the complexity of the attacks.

Adversary detection systems fail in the real-time classification of attacks due to the speed of the connection. Attack detection and defense become less accurate when more data is involved, increasing the number of false positive attack detections. This, in turn, negatively affects legitimate data. The priority of researchers, therefore, should be to see how they can solve the trade-off between accuracy and cost, and ensure that both performance and adversarial robustness are achieved.

B. FUTURE DIRECTION

1. Development of Multi-layered Defense Mechanisms

In future studies, there is a need to develop multi-layered defense mechanisms for Network Intrusion Detection Systems (NIDS) to strengthen their resilience against adversarial attacks. This involves integrating of multiple coordinated defense strategies, which involves the combination of anomaly detection with signature-based methods and the introduction of machine learning models that can adapt to the evolving tactics of attacks. By combining these layers of defense, researchers will improve the resiliency of NIDS by adopting a comprehensive and adaptive approach, which will counter threats posed by adversaries in cyber security.

2. Lightweight Adaptive Attack-Resistant Strategies

Future research should focus on developing lightweight, smart, and adaptable defense mechanisms for the organization of Network Intrusion Detection Systems (NIDS) able to

withstand adversarial attacks. Emphasizing resource-efficient solutions that do not deteriorate detection accuracy, as well as the development of lightweight machine learning and real-time adaptive methods are important. The efficiency in addressing the challenge of adversarial attacks in NIDS by considering this aspect will shape the enhanced agility and responsiveness of such types of systems in the new cybersecurity domain.

3. Universal Defense and Adaptive Attack Strategies

Future research should focus on developing universal technique resistant to adaptive attacks on Network Intrusion Detection Systems (NIDS). This means looking at cross-domain defense techniques which can be adapted to different attack scenarios. Researchers can aid in the creation of NIDS frameworks that are flexible and robust against the unique and ever-changing threat tactics in various network environments by focusing on a holistic and versatile defense strategy.

4. Application of Transfer Learning for Improved Detection Accuracy

To improve the detection accuracy of Network Intrusion Detection systems (NIDS) under adversarial attacks, future studies should focus on using transfer learning. Applying knowledge gained from one domain for the enhancement of detection capacities in another can give a defense against ever-changing adversarial strategies. Investigating optimal transfer learning methods and their adaptation to NIDS scenarios holds great promise for advancing intrusion detection systems for resilience in complex and dynamic network environments.

VI. CONCLUSION

This research paper analyses various adversarial attacks, defense techniques for Network Intrusion Detection Systems (NIDS), benchmark datasets, and commercial intrusion detection system (IDS) tools. It also demonstrates the classification of adversarial attacks using deep learning algorithms. The findings reveal that the application space of deep learning-based algorithms in the domain of NIDS is limited due to their impracticality. In contrast, GAN-based and packet-based attacks are more viable and serve as a benchmark for implementable solutions. Packet-based approaches, in particular, are more practical than GAN. The study also finds that the packet-mutation approach is an effective method to preserve the robustness of deep learning networks against adversarial attacks. However, the number of studies validating this approach is limited. Additionally, despite active research in this field, IDS has not been widely adopted in business, although some companies have traditionally used IDS in their operations. The paper concludes by calling for more research and innovation in cybersecurity to effectively outmaneuver the continuing evolution of adversarial threats and ensure the security of the digital environment. The study highlights the need to explore novel approaches to counter adversarial attacks and promote the widespread adoption of IDS in the business domain.

VII. ACKNOWLEDGMENT

This research was supported by the Connectivity Innovation Network (CIN) under project ID 104144. We sincerely thank CIN for their financial support, which made this study possible. We also acknowledge the valuable contributions of our colleagues and the support from our institution, Charles Sturt University.

REFERENCES

- [1] J. Gubbi, R. Buyya, S. Marusic, and M. Palaniswami, "Internet of things (IoT): A vision, architectural elements, and future directions," *Future generation computer systems*, vol. 29, no. 7, pp. 1645–1660, 2013.
- [2] N. Virvilis and D. Gritzalis, "The big four-what we did wrong in advanced persistent threat detection?," in 2013 international conference on availability, reliability and security, pp. 248–254, IEEE, 2013.
- [3] K. Scarfone, P. Mell, et al., "Guide to intrusion detection and prevention systems (IDPS)," NIST special publication, vol. 800, no. 2007, p. 94, 2007.
- [4] I. Corona, G. Giacinto, and F. Roli, "Adversarial attacks against intrusion detection systems: Taxonomy, solutions and open issues," *Information Sciences*, vol. 239, pp. 201–225, 2013.
- [5] D. Mutz, C. Kruegel, W. Robertson, G. Vigna, and R. Kemmerer, "Reverse engineering of network signatures," in *Proceedings of the Information Technology Security Conference*, 2005.
- [6] Z. Ahmad, A. Shahid Khan, C. Wai Shiang, J. Abdullah, and F. Ahmad, "Network intrusion detection system: A systematic study of machine learning and deep learning approaches," *Transactions on Emerging Telecommunications Technologies*, vol. 32, no. 1, p. e4150, 2021.
- [7] O. Ibitoye, R. Abou-Khamis, A. Matrawy, and M. O. Shafiq, "The threat of adversarial attacks on machine learning in network security—a survey," *arXiv preprint arXiv:1911.02621*, 2019.
- [8] A. L. Buczak and E. Guven, "A survey of data mining and machine learning methods for cyber security intrusion detection," *IEEE Communications surveys & tutorials*, vol. 18, no. 2, pp. 1153–1176, 2015.
- [9] N. Chaabouni, M. Mosbah, A. Zemhari, C. Sauvignac, and P. Faruki, "Network intrusion detection for IoT security based on learning techniques," *IEEE Communications Surveys & Tutorials*, vol. 21, no. 3, pp. 2671–2701, 2019.
- [10] F. Hussain, R. Hussain, S. A. Hassan, and E. Hossain, "Machine learning in IoT security: Current solutions and future challenges," *IEEE Communications Surveys & Tutorials*, vol. 22, no. 3, pp. 1686–1721, 2020.
- [11] C. Zhang, P. Patras, and H. Haddadi, "Deep learning in mobile and wireless networking: A survey," *IEEE Communications surveys & tutorials*, vol. 21, no. 3, pp. 2224–2287, 2019.
- [12] P. Stegeby, "Intrusion detection systems: A comparison in configuration and implementation between ossec and snort," 2023.
- [13] X. Wang, J. Li, X. Kuang, Y.-a. Tan, and J. Li, "The security of machine learning in an adversarial setting: A survey," *Journal of Parallel and Distributed Computing*, vol. 130, pp. 12–23, 2019.
- [14] G. Li, P. Zhu, J. Li, Z. Yang, N. Cao, and Z. Chen, "Security matters: A survey on adversarial machine learning," *arXiv preprint arXiv:1810.07339*, 2018.
- [15] A. Chakraborty, M. Alam, V. Dey, A. Chattopadhyay, and D. Mukhopadhyay, "Adversarial attacks and defences: A survey," *arXiv preprint arXiv:1810.00069*, 2018.
- [16] S. H. Silva and P. Najafirad, "Opportunities and challenges in deep learning adversarial robustness: A survey," *arXiv preprint arXiv:2007.00753*, 2020.
- [17] A. Serban, E. Poll, and J. Visser, "Adversarial examples on object recognition: A comprehensive survey," *ACM Computing Surveys (CSUR)*, vol. 53, no. 3, pp. 1–38, 2020.
- [18] S. Bhambri, S. Muku, A. Tulasi, and A. B. Buduru, "A study of black box adversarial attacks in computer vision," *arXiv preprint arXiv:1912.01667*, 2019.
- [19] W. E. Zhang, Q. Z. Sheng, A. Alhazmi, and C. Li, "Adversarial attacks on deep-learning models in natural language processing: A survey," *ACM Transactions on Intelligent Systems and Technology (TIST)*, vol. 11, no. 3, pp. 1–41, 2020.
- [20] F. O. Olowononi, D. B. Rawat, and C. Liu, "Resilient machine learning for networked cyber physical systems: A survey for machine learning security to securing machine learning for CPS," *IEEE Communications Surveys & Tutorials*, vol. 23, no. 1, pp. 524–552, 2020.
- [21] N. Martins, J. M. Cruz, T. Cruz, and P. H. Abreu, "Adversarial machine learning applied to intrusion and malware scenarios: a systematic review," *IEEE Access*, vol. 8, pp. 35403–35419, 2020.
- [22] O. Ibitoye, R. Abou-Khamis, A. Matrawy, and M. O. Shafiq, "The threat of adversarial attacks on machine learning in network security—a survey," *arXiv preprint arXiv:1911.02621*, 2019.
- [23] H. A. Alatwi and C. Morisset, "Adversarial machine learning in network intrusion detection domain: A systematic review," *arXiv preprint arXiv:2112.03315*, 2021.
- [24] Y. Wang, J. Liu, X. Chang, J. Wang, and R. J. Rodríguez, "AB-FGSM: Adabelief optimizer and FGSM-based approach to generate adversarial examples," *Journal of Information Security and Applications*, vol. 68, 2022.
- [25] Z. Leng, Z. Cheng, P. Wei, and J. Chen, "Object-aware transfer-based black-box adversarial attack on object detector," *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, vol. 14436 LNCS, p. 278 – 289, 2024.
- [26] Y. Aiza, C. Zhang, and J. Yu, "Black-box targeted adversarial attack based on multi-population genetic algorithm," in 2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC), pp. 317–322, 2023.
- [27] B. Vivek, K. R. Mopuri, and R. V. Babu, "Gray-box adversarial training," in *Proceedings of the European conference on computer vision (ECCV)*, pp. 203–218, 2018.
- [28] Y. Khare, K. Lakara, S. Mittal, A. Kaushik, and R. Singhal, "SpotOn: A gradient-based targeted data poisoning attack on deep neural networks," vol. 2023-April, 2023.
- [29] M. Li and C. Cao, "Defense against adversarial attacks using image label and pixel guided sparse denoiser," in 2022 7th International Conference on Big Data Analytics (ICBDA), pp. 253–258, 2022.
- [30] M. N. Al-Andoli, S. C. Tan, K. S. Sim, P. Y. Goh, and C. P. Lim, "A framework for robust deep learning models against adversarial attacks based on a protection layer approach," *IEEE Access*, pp. 1–1, 2024.
- [31] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," *arXiv preprint arXiv:1412.6572*, 2014.
- [32] N. Carlini and D. Wagner, "Towards evaluating the robustness of neural networks," in 2017 IEEE Symposium on Security and Privacy (SP), pp. 39–57, IEEE, 2017.
- [33] B. Kim, Y. E. Sagduyu, K. Davaslioglu, T. Erpek, and S. Ulukus, "Channel-aware adversarial attacks against deep learning-based wireless signal classifiers," *IEEE Transactions on Wireless Communications*, vol. 21, no. 6, pp. 3868–3880, 2021.
- [34] Y. Mirsky and W. Lee, "The creation and detection of DeepFakes: A survey," *ACM Computing Surveys (CSUR)*, vol. 54, no. 1, pp. 1–41, 2021.
- [35] Y. Ye, Y. Chen, and M. Liu, "Multiuser adversarial attack on deep learning for OFDM detection," *IEEE Wireless Communications Letters*, vol. 11, no. 12, pp. 2527–2531, 2022.
- [36] P. Papadopoulos, O. Thornevill von Essen, N. Pitropakis, C. Chrysoulas, A. Mylonas, and W. J. Buchanan, "Launching adversarial attacks against network intrusion detection systems for IoT," *Journal of Cybersecurity and Privacy*, vol. 1, no. 2, pp. 252–273, 2021.
- [37] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," *arXiv preprint arXiv:1706.06083*, 2017.
- [38] F. Xiao, Y. Huang, Y. Zuo, W. Kuang, and W. Wang, "Over-The-Air adversarial attacks on deep learning Wi-Fi fingerprinting," *IEEE Internet of Things Journal*, 2023.
- [39] A. Ilioudi, A. Dabiri, B. J. Wolf, and B. De Schutter, "Deep learning for object detection and segmentation in videos: Toward an integration with domain knowledge," *IEEE Access*, vol. 10, pp. 34562–34576, 2022.
- [40] R. Duan, X. Mao, A. K. Qin, Y. Chen, S. Ye, Y. He, and Y. Yang, "Adversarial laser beam: Effective physical-world attack to DNNs in a blink," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 16062–16071, 2021.
- [41] H. Hirano, A. Minagi, and K. Takemoto, "Universal adversarial attacks on deep neural networks for medical image classification," *BMC medical imaging*, vol. 21, pp. 1–13, 2021.
- [42] T. K. Tasooji and H. J. Marquez, "A secure decentralized event-triggered cooperative localization in multi-robot systems under cyber attack," *IEEE Access*, vol. 10, pp. 128101–128121, 2022.

- [43] Y.-Y. Zhang, M.-Y. Huang, and X. Zhang, "Occupation mechanism for eliminating double-spending attacks on trusted transaction blockchain," in 2021 The 3rd International Conference on Blockchain Technology, pp. 14–19, 2021.
- [44] H. Badrsimaei, R.-A. Hooshmand, and S. Nobakhtian, "Stealthy and profitable data injection attack on real time electricity market with network model uncertainties," *Electric Power Systems Research*, vol. 205, p. 107742, 2022.
- [45] G. Gianini, L. G. Fossi, C. Mio, O. Caelen, L. Brunie, and E. Damiani, "Managing a pool of rules for credit card fraud detection by a game theory based approach," *Future Generation Computer Systems*, vol. 102, pp. 549–561, 2020.
- [46] S. Guo, J. Zhao, X. Li, J. Duan, D. Mu, and X. Jing, "A black-box attack method against machine-learning-based anomaly network flow detection models," *Security and Communication Networks*, vol. 2021, pp. 1–13, 2021.
- [47] Y. Pacheco and W. Sun, "Adversarial machine learning: A comparative study on contemporary intrusion detection datasets," in *ICISSP*, pp. 160–171, 2021.
- [48] Z. Wang, "Deep learning-based intrusion detection with adversaries," *IEEE Access*, vol. 6, pp. 38367–38384, 2018.
- [49] B.-E. Zolbayer, R. Sheatsley, P. McDaniel, M. J. Weisman, S. Zhu, S. Zhu, and S. Krishnamurthy, "Generating practical adversarial network traffic flows using NIDSGAN," arXiv preprint arXiv:2203.06694, 2022.
- [50] W. Ford, "Chapter 20 - basic iterative methods," in *Numerical Linear Algebra with Applications* (W. Ford, ed.), pp. 469–490, Boston: Academic Press, 2015.
- [51] H. Liu, Z. Ge, Z. Zhou, F. Shang, Y. Liu, and L. Jiao, "Gradient correction for white-box adversarial attacks," *IEEE Transactions on Neural Networks and Learning Systems*, pp. 1–12, 2023.
- [52] E. Nowroozi, Y. Mekdad, M. H. Berenjestanaki, M. Conti, and A. E. Fergougui, "Demystifying the transferability of adversarial attacks in computer networks," *IEEE Transactions on Network and Service Management*, vol. 19, no. 3, p. 3387–3400, 2022.
- [53] N. Papernot, P. McDaniel, S. Jha, M. Fredrikson, Z. B. Celik, and A. Swami, "The limitations of deep learning in adversarial settings," in 2016 IEEE European symposium on security and privacy (EuroS&P), pp. 372–387, IEEE, 2016.
- [54] S.-M. Moosavi-Dezfooli, A. Fawzi, and P. Frossard, "DeepFool: a simple and accurate method to fool deep neural networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 2574–2582, 2016.
- [55] Y. Xu, X. Zhong, A. J. Yepes, and J. H. Lau, "Grey-box adversarial attack and defence for sentiment classification," arXiv preprint arXiv:2103.11576, 2021.
- [56] Z. Che, A. Borji, G. Zhai, S. Ling, J. Li, Y. Tian, G. Guo, and P. Le Callet, "Adversarial attack against deep saliency models powered by non-redundant priors," *IEEE Transactions on Image Processing*, vol. 30, pp. 1973–1988, 2021.
- [57] L. Schwinn, A. Nguyen, R. Raab, D. Zanca, B. M. Eskofier, D. Tenbrinck, and M. Burger, "Dynamically sampled nonlocal gradients for stronger adversarial attacks," in 2021 International Joint Conference on Neural Networks (IJCNN), pp. 1–8, IEEE, 2021.
- [58] Y. Yu, X. Gao, and C.-Z. Xu, "LAFEAT: Piercing through adversarial defenses with latent features," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 5735–5745, 2021.
- [59] Y. Xiang, Y. Xu, Y. Li, W. Ma, Q. Xuan, and Y. Liu, "Side-channel gray-box attack for DNNs," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 68, no. 1, pp. 501–505, 2020.
- [60] H. Wang, S. Wang, Z. Jin, Y. Wang, C. Chen, and M. Tistarelli, "Similarity-based gray-box adversarial attack against deep face recognition," in 2021 16th IEEE international conference on automatic face and gesture recognition (FG 2021), pp. 1–8, IEEE, 2021.
- [61] S. Chen, Z. He, C. Sun, J. Yang, and X. Huang, "Universal adversarial attack on attention and the resulting dataset damagenet," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 4, pp. 2188–2197, 2020.
- [62] Z. Liu, Y. Luo, L. Wu, S. Li, Z. Liu, and S. Z. Li, "Are gradients on graph structure reliable in gray-box attacks?," in *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*, pp. 1360–1368, 2022.
- [63] J. Li, T. Xie, L. Chen, F. Xie, X. He, and Z. Zheng, "Adversarial attack on large scale graph," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 1, pp. 82–95, 2021.
- [64] E. D. Cubuk, B. Zoph, S. S. Schoenholz, and Q. V. Le, "Intriguing properties of adversarial examples," arXiv preprint arXiv:1711.02846, 2017.
- [65] S. Chen, Z. Chen, and D. Wang, "Adaptive adversarial training for meta reinforcement learning," in 2021 International Joint Conference on Neural Networks (IJCNN), pp. 1–8, IEEE, 2021.
- [66] S. Feng, F. Feng, X. Xu, Z. Wang, Y. Hu, and L. Xie, "Digital watermark perturbation for adversarial examples to fool deep neural networks," in 2021 International Joint Conference on Neural Networks (IJCNN), pp. 1–8, IEEE, 2021.
- [67] W. Wu, Y. Su, M. R. Lyu, and I. King, "Improving the transferability of adversarial samples with adversarial transformations," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 9024–9033, 2021.
- [68] C. Luo, Q. Lin, W. Xie, B. Wu, J. Xie, and L. Shen, "Frequency-driven imperceptible adversarial attack on semantic similarity," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 15315–15324, 2022.
- [69] H. Chang, Y. Rong, T. Xu, W. Huang, H. Zhang, P. Cui, X. Wang, W. Zhu, and J. Huang, "Adversarial attack framework on graph embedding models with limited knowledge," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 5, pp. 4499–4513, 2022.
- [70] H. Cheng, H. Liu, F. Gao, and Z. Chen, "Adgan: A scalable gan-based architecture for image anomaly detection," in 2020 IEEE 4th Information Technology, Networking, Electronic and Automation Control Conference (ITNEC), vol. 1, pp. 987–993, IEEE, 2020.
- [71] Q. Li, Y. Qi, Q. Hu, S. Qi, Y. Lin, and J. S. Dong, "Adversarial adaptive neighborhood with feature importance-aware convex interpolation," *IEEE Transactions on Information Forensics and Security*, vol. 16, pp. 2447–2460, 2020.
- [72] R. Duan, X. Mao, A. K. Qin, Y. Chen, S. Ye, Y. He, and Y. Yang, "Adversarial laser beam: Effective physical-world attack to dnns in a blink," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 16062–16071, 2021.
- [73] R. Duan, X. Ma, Y. Wang, J. Bailey, A. K. Qin, and Y. Yang, "Adversarial camouflage: Hiding physical-world attacks with natural styles," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 1000–1008, 2020.
- [74] E. Yang, T. Liu, C. Deng, and D. Tao, "Adversarial examples for hamming space search," *IEEE transactions on cybernetics*, vol. 50, no. 4, pp. 1473–1484, 2018.
- [75] R. Duan, Y. Chen, D. Niu, Y. Yang, A. K. Qin, and Y. He, "Advdrop: Adversarial attack to dnns by dropping information," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 7506–7515, 2021.
- [76] H. Wang, S. Wang, Z. Jin, Y. Wang, C. Chen, and M. Tistarelli, "Similarity-based gray-box adversarial attack against deep face recognition," in 2021 16th IEEE international conference on automatic face and gesture recognition (FG 2021), pp. 1–8, IEEE, 2021.
- [77] R. Vinayakumar, M. Alazab, K. Soman, P. Poornachandran, A. Al-Nemrat, and S. Venkatraman, "Deep learning approach for intelligent intrusion detection system," *IEEE Access*, vol. 7, pp. 41525–41550, 2019.
- [78] S. Jia, Y. Song, C. Ma, and X. Yang, "IoU attack: Towards temporally coherent black-box adversarial attack for visual object tracking," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 6709–6718, 2021.
- [79] J. Chen, M. I. Jordan, and M. J. Wainwright, "HopSkipJumpAttack: A query-efficient decision-based attack," in 2020 IEEE symposium on security and privacy (sp), pp. 1277–1294, IEEE, 2020.
- [80] M. Cheng, T. Le, P.-Y. Chen, J. Yi, H. Zhang, and C.-J. Hsieh, "Query-efficient hard-label black-box attack: An optimization-based approach," arXiv preprint arXiv:1807.04457, 2018.
- [81] J. Chen, M. Su, S. Shen, H. Xiong, and H. Zheng, "POBA-GA: Perturbation optimized black-box adversarial attacks via genetic algorithm," *Computers & Security*, vol. 85, pp. 89–106, 2019.
- [82] Y. Zhang, X. Tian, Y. Li, X. Wang, and D. Tao, "Principal component adversarial example," *IEEE Transactions on Image Processing*, vol. 29, pp. 4804–4815, 2020.
- [83] F. Croce and M. Hein, "Minimally distorted adversarial examples with a fast adaptive boundary attack," in *International Conference on Machine Learning*, pp. 2196–2205, PMLR, 2020.
- [84] W. Huang, X. Peng, Z. Shi, and Y. Ma, "Adversarial attack against LSTM-based DDoS intrusion detection system," in 2020 IEEE 32nd

- International Conference on Tools with Artificial Intelligence (ICTAI), pp. 686–693, IEEE, 2020.
- [85] J. Yang, T. Li, G. Liang, W. He, and Y. Zhao, “A simple recurrent unit model based intrusion detection system with DCGAN,” *IEEE Access*, vol. 7, pp. 83286–83296, 2019.
- [86] S. P. Kulyadi, P. Mohandas, S. K. S. Kumar, M. S. Raman, and V. Vasan, “Anomaly detection using generative adversarial networks on firewall log message data,” in 2021 13th International Conference on Electronics, Computers and Artificial Intelligence (ECAI), pp. 1–6, IEEE, 2021.
- [87] M. Usama, M. Asim, S. Latif, J. Qadir, et al., “Generative adversarial networks for launching and thwarting adversarial attacks on network intrusion detection systems,” in 2019 15th international wireless communications & mobile computing conference (IWCMC), pp. 78–83, IEEE, 2019.
- [88] C. Choi, S. Shin, and I. Lee, “Opcode sequence amplifier using sequence generative adversarial networks,” in 2019 International Conference on Information and Communication Technology Convergence (ICTC), pp. 968–970, IEEE, 2019.
- [89] Y. Liu, J. Li, B. Liu, X. Gao, and X. Liu, “Malware identification method based on image analysis,” in 2021 11th International Conference on Information Technology in Medicine and Education (ITME), pp. 157–161, IEEE, 2021.
- [90] X. Peng, H. Xian, Q. Lu, and X. Lu, “Semantics aware adversarial malware examples generation for black-box attacks,” *Applied Soft Computing*, vol. 109, p. 107506, 2021.
- [91] W. L. Tan and T. Truong-Huu, “Enhancing robustness of malware detection using synthetically-adversarial samples,” in GLOBECOM 2020-2020 IEEE Global Communications Conference, pp. 1–6, IEEE, 2020.
- [92] J.-T. Wang and C.-H. Wang, “High performance WGAN-GP based multiple-category network anomaly classification system,” in 2019 International Conference on Cyber Security for Emerging Technologies (CSET), pp. 1–7, IEEE, 2019.
- [93] Q. Yan, M. Wang, W. Huang, X. Luo, and F. R. Yu, “Automatically synthesizing DoS attack traces using generative adversarial networks,” *International journal of machine learning and cybernetics*, vol. 10, no. 12, pp. 3387–3396, 2019.
- [94] Z. Wang, P. Wang, X. Zhou, S. Li, and M. Zhang, “FLOWGAN: Unbalanced network encrypted traffic identification method based on GAN,” in 2019 IEEE Intl Conf on Parallel & Distributed Processing with Applications, Big Data & Cloud Computing, Sustainable Computing & Communications, Social Computing & Networking (ISPA/BDCLOUD/SocialCom/SustainCom), pp. 975–983, IEEE, 2019.
- [95] J. Lee and K. Park, “AE-CGAN model based high performance network intrusion detection system,” *Applied Sciences*, vol. 9, no. 20, p. 4221, 2019.
- [96] S. Liu, A. Shrivastava, J. Du, and L. Zhong, “Better accuracy with quantified privacy: representations learned via reconstructive adversarial network,” *arXiv preprint arXiv:1901.08730*, 2019.
- [97] S. Zhao, J. Li, J. Wang, Z. Zhang, L. Zhu, and Y. Zhang, “attackGAN: Adversarial attack against black-box ids using generative adversarial networks,” *Procedia Computer Science*, vol. 187, pp. 128–133, 2021.
- [98] B. Nugraha, N. Kulkarni, and A. Gopikrishnan, “Detecting adversarial DDoS attacks in software-defined networking using deep learning techniques and adversarial training,” in 2021 IEEE International Conference on Cyber Security and Resilience (CSR), pp. 448–454, IEEE, 2021.
- [99] T. Zixu, K. S. K. Liyanage, and M. Gurusamy, “Generative adversarial network and auto encoder based anomaly detection in distributed IoT networks,” in GLOBECOM 2020-2020 IEEE Global Communications Conference, pp. 1–7, IEEE, 2020.
- [100] S. M. W. Umba, A. M. Abu-Mahfouz, T. Ramotsoela, and G. P. Hancke, “A review of artificial intelligence based intrusion detection for software-defined wireless sensor networks,” in 2019 IEEE 28th International symposium on industrial electronics (ISIE), pp. 1277–1282, IEEE, 2019.
- [101] P. Radoglou Grammatikis, P. Sarigiannidis, G. Efstathopoulos, and E. Panaousis, “ARIES: A novel multivariate intrusion detection system for smart grid,” *Sensors*, vol. 20, no. 18, p. 5305, 2020.
- [102] K. Kim, J. S. Kim, S. Jeong, J.-H. Park, and H. K. Kim, “Cybersecurity for autonomous vehicles: Review of attacks and defense,” *Computers & Security*, vol. 103, p. 102150, 2021.
- [103] E. Seo, H. M. Song, and H. K. Kim, “GIDS: GAN based intrusion detection system for in-vehicle network,” in 2018 16th Annual Conference on Privacy, Security and Trust (PST), pp. 1–6, IEEE, 2018.
- [104] H. Sedjelmaci, “Attacks detection and decision framework based on generative adversarial network approach: Case of vehicular edge computing network,” *Transactions on Emerging Telecommunications Technologies*, vol. 33, no. 10, p. e4073, 2022.
- [105] Z. Lin, Y. Shi, and Z. Xue, “IDSGAN: Generative adversarial networks for attack generation against intrusion detection,” in Pacific-asia conference on knowledge discovery and data mining, pp. 79–91, Springer, 2022.
- [106] S. Wang, Q. Wang, Z. Jiang, X. Wang, and R. Jing, “A weak coupling of semi-supervised learning with generative adversarial networks for malware classification,” in 2020 25th International Conference on Pattern Recognition (ICPR), pp. 3775–3782, IEEE, 2021.
- [107] C. Sekhar, K. V. Rao, and M. H. M. K. Prasad, “Classification performance improvement by enhancing the detection accuracy of DDoS attacks over flash crowd using CROSS GAN (XGAN),” *Multimedia Tools and Applications*, vol. 82, no. 25, p. 38693 – 38714, 2023. Cited by: 1.
- [108] N. Zhu, G. Zhao, Y. Yang, H. Yang, and Z. Liu, “AEC_GAN: Unbalanced data processing decision-making in network attacks based on ACGAN and machine learning,” *IEEE Access*, vol. 11, p. 52452 – 52465, 2023. Cited by: 1; All Open Access, Gold Open Access.
- [109] A.-G. Mari, D. Zinca, and V. Dobrota, “Development of a machine-learning intrusion detection system and testing of its performance using a generative adversarial network,” *Sensors*, vol. 23, no. 3, 2023. Cited by: 4; All Open Access, Gold Open Access, Green Open Access.
- [110] V. Kumar and D. Sinha, “Synthetic attack data generation model applying generative adversarial network for intrusion detection,” *Computers and Security*, vol. 125, 2023. Cited by: 7.
- [111] Z. Li, S. Chen, H. Dai, D. Xu, C.-K. Chu, and B. Xiao, “Abnormal traffic detection: Traffic feature extraction and DAE-GAN with efficient data augmentation,” *IEEE Transactions on Reliability*, vol. 72, no. 2, p. 498 – 510, 2023. Cited by: 2.
- [112] K. T. Chui, B. B. Gupta, P. Chaurasia, V. Arya, A. Almomani, and W. Al-halabi, “Three-stage data generation algorithm for multiclass network intrusion detection with highly imbalanced dataset,” *International Journal of Intelligent Networks*, vol. 4, p. 202 – 210, 2023. Cited by: 0; All Open Access, Gold Open Access.
- [113] W. Ding, “A GAN-based security strategy for WSN networks based on honeypot algorithm,” *Physical Communication*, vol. 62, p. 102260, 2024.
- [114] A. Ferdowsi and W. Saad, “Generative adversarial networks for distributed intrusion detection in the Internet of things,” in 2019 IEEE Global Communications Conference (GLOBECOM), pp. 1–6, IEEE, 2019.
- [115] N. Abdalgawad, A. Sajun, Y. Kaddoura, I. A. Zualkernan, and F. Aloul, “Generative deep learning to detect cyberattacks for the IoT-23 dataset,” *IEEE Access*, vol. 10, pp. 6430–6441, 2021.
- [116] H. Zhang, I. Goodfellow, D. Metaxas, and A. Odena, “Self-attention generative adversarial networks,” in International conference on machine learning, pp. 7354–7363, PMLR, 2019.
- [117] S. Li, Y. Cao, S. Liu, Y. Lai, Y. Zhu, and N. Ahmad, “HDA-IDS: A hybrid DoS attacks intrusion detection system for IoT by using semi-supervised cl-gan,” *Expert Systems with Applications*, vol. 238, p. 122198, 2024.
- [118] E. Ntizikira, L. Wang, J. Chen, and K. Saleem, “Honey-block: Edge assisted ensemble learning model for intrusion detection and prevention using defense mechanism in IoT,” *Computer Communications*, vol. 214, p. 1 – 17, 2024. Cited by: 0.
- [119] Y. Wu, L. Nie, S. Wang, Z. Ning, and S. Li, “Intelligent intrusion detection for Internet of Things security: A deep convolutional generative adversarial network-enabled approach,” *IEEE Internet of Things Journal*, vol. 10, no. 4, p. 3094 – 3106, 2023. Cited by: 13.
- [120] Z. Liu, J. Hu, Y. Liu, K. Roy, X. Yuan, and J. Xu, “Anomaly-based intrusion on IoT networks using AIGAN — a generative adversarial network,” *IEEE Access*, vol. 11, p. 91116 – 91132, 2023. Cited by: 0; All Open Access, Gold Open Access.
- [121] S. Kably, T. Benbarrad, N. Alaoui, and M. Arioua, “Multi-zone-wise blockchain based intrusion detection and prevention system for IoT environment,” *Computers, Materials and Continua*, vol. 74, no. 1, p. 253 – 278, 2023. Cited by: 3; All Open Access, Gold Open Access.
- [122] M. Poongodi and M. Hamdi, “Intrusion detection system using distributed multilevel discriminator in GAN for IoT system,” *Transactions on Emerging Telecommunications Technologies*, vol. 34, no. 11, 2023. Cited by: 0; All Open Access, Hybrid Gold Open Access.
- [123] C. Park, J. Lee, Y. Kim, J.-G. Park, H. Kim, and D. Hong, “An enhanced AI-based network intrusion detection system using generative adversarial networks,” *IEEE Internet of Things Journal*, vol. 10, no. 3, p. 2330 – 2345, 2023. Cited by: 20; All Open Access, Hybrid Gold Open Access.

- [124] P. F. De Araujo-Filho, M. Naili, G. Kaddoum, E. T. Fapi, and Z. Zhu, "Unsupervised GAN-based intrusion detection system using temporal convolutional networks and self-attention," *IEEE Transactions on Network and Service Management*, vol. 20, no. 4, p. 4951 – 4963, 2023. Cited by: 3.
- [125] B. Sharma, L. Sharma, C. Lal, and S. Roy, "Anomaly based network intrusion detection for IoT attacks using deep learning technique," *Computers and Electrical Engineering*, vol. 107, 2023. Cited by: 18; All Open Access, Green Open Access.
- [126] C. Zhang, P. Patras, and H. Haddadi, "Deep learning in mobile and wireless networking: A survey," *IEEE Communications surveys & tutorials*, vol. 21, no. 3, pp. 2224–2287, 2019.
- [127] C. Zhang, X. Ouyang, and P. Patras, "ZipNet-GAN: Inferring fine-grained mobile traffic patterns via a generative adversarial neural network," in *Proceedings of the 13th International Conference on emerging Networking EXperiments and Technologies*, pp. 363–375, 2017.
- [128] S. M. W. Umba, A. M. Abu-Mahfouz, T. Ramotsoela, and G. P. Hancke, "A review of artificial intelligence based intrusion detection for software-defined wireless sensor networks," in *2019 IEEE 28th International symposium on industrial electronics (ISIE)*, pp. 1277–1282, IEEE, 2019.
- [129] P. Radoglou Grammatikis, P. Sarigiannidis, G. Efstathopoulos, and E. Panaousis, "ARIES: A novel multivariate intrusion detection system for smart grid," *Sensors*, vol. 20, no. 18, p. 5305, 2020.
- [130] S. Rajasoundaran, S. V. N. S. Kumar, M. Selvi, K. Thangaramya, and K. Arputharaj, "Secure and optimized intrusion detection scheme using LSTM-MAC principles for underwater wireless sensor networks," *Wireless Networks*, vol. 30, no. 1, p. 209 – 231, 2024. Cited by: 2.
- [131] Z. Cai, A. Wang, W. Zhang, M. Gruffke, and H. Schweppe, "0-days & mitigations: roadways to exploit and secure connected BMW cars," *Black Hat USA*, vol. 2019, no. 39, p. 6, 2019.
- [132] H. Qiu, T. Dong, T. Zhang, J. Lu, G. Memmi, and M. Qiu, "Adversarial attacks against network intrusion detection in IoT systems," *IEEE Internet of Things Journal*, vol. 8, no. 13, pp. 10327–10335, 2020.
- [133] A. Ilyas, L. Engstrom, A. Athalye, and J. Lin, "Black-box adversarial attacks with limited queries and information," in *International conference on machine learning*, pp. 2137–2146, PMLR, 2018.
- [134] E. Alhajjar, P. Maxwell, and N. Bastian, "Adversarial machine learning in network intrusion detection systems," *Expert Systems with Applications*, vol. 186, p. 115782, 2021.
- [135] A. Kuppa, S. Grzonkowski, M. R. Asghar, and N.-A. Le-Khac, "Black box attacks on deep anomaly detectors," in *Proceedings of the 14th international conference on availability, reliability and security*, pp. 1–10, 2019.
- [136] M. Ali, Y.-F. Hu, D. K. Luong, G. Oguntala, J.-P. Li, and K. Abdo, "Adversarial attacks on ai based intrusion detection system for heterogeneous wireless communications networks," in *2020 AIAA/IEEE 39th Digital Avionics Systems Conference (DASC)*, pp. 1–6, IEEE, 2020.
- [137] W. Huang, X. Peng, Z. Shi, and Y. Ma, "Adversarial attack against LSTM-based DDoS intrusion detection system," in *2020 IEEE 32nd International Conference on Tools with Artificial Intelligence (ICTAI)*, pp. 686–693, IEEE, 2020.
- [138] A. Kuppa, S. Grzonkowski, M. R. Asghar, and N.-A. Le-Khac, "Black box attacks on deep anomaly detectors," in *Proceedings of the 14th international conference on availability, reliability and security*, pp. 1–10, 2019.
- [139] T. Fladby, H. Haugerud, S. Nichele, K. Begnum, and A. Yazidi, "Evading a machine learning-based intrusion detection system through adversarial perturbations," in *Proceedings of the International Conference on Research in Adaptive and Convergent Systems*, pp. 161–166, 2020.
- [140] W. Brendel, J. Rauber, and M. Bethge, "Decision-based adversarial attacks: Reliable attacks against black-box machine learning models," *arXiv preprint arXiv:1712.04248*, 2017.
- [141] G. Apruzzese, M. Colajanni, L. Ferretti, and M. Marchetti, "Addressing adversarial attacks against security systems based on machine learning," in *2019 11th international conference on cyber conflict (CyCon)*, vol. 900, pp. 1–18, IEEE, 2019.
- [142] L. Schott, J. Rauber, M. Bethge, and W. Brendel, "Towards the first adversarially robust neural network model on mnist," *arXiv preprint arXiv:1805.09190*, 2018.
- [143] K. Yang, J. Liu, C. Zhang, and Y. Fang, "Adversarial examples against the deep learning based network intrusion detection systems," in *MILCOM 2018-2018 IEEE military communications conference (MILCOM)*, pp. 559–564, IEEE, 2018.
- [144] Y. Sharon, D. Berend, Y. Liu, A. Shabtai, and Y. Elovici, "TANTRA: Timing-based adversarial network traffic reshaping attack," *IEEE Transactions on Information Forensics and Security*, vol. 17, pp. 3225–3237, 2022.
- [145] D. Han, Z. Wang, Y. Zhong, W. Chen, J. Yang, S. Lu, X. Shi, and X. Yin, "Practical traffic-space adversarial attacks on learning-based NIDSs," 2020.
- [146] I. Homoliak, M. Teknos, M. Ochoa, D. Breitenbacher, S. Hosseini, and P. Hanacek, "Improving network intrusion detection classifiers by non-payload-based exploit-independent obfuscations: An adversarial approach," *arXiv preprint arXiv:1805.02684*, 2018.
- [147] M. J. Hashemi, G. Cusack, and E. Keller, "Towards evaluation of NIDSs in adversarial setting," in *Proceedings of the 3rd ACM CoNEXT Workshop on Big Data, Machine Learning and Artificial Intelligence for Data Communication Networks*, pp. 14–21, 2019.
- [148] B. Zong, Q. Song, M. R. Min, W. Cheng, C. Lumezanu, D. Cho, and H. Chen, "Deep autoencoding Gaussian mixture model for unsupervised anomaly detection," in *International conference on learning representations*, 2018.
- [149] A. Kuppa, S. Grzonkowski, M. R. Asghar, and N.-A. Le-Khac, "Black box attacks on deep anomaly detectors," in *Proceedings of the 14th international conference on availability, reliability and security*, pp. 1–10, 2019.
- [150] Y. Mirsky, T. Doitshman, Y. Elovici, and A. Shabtai, "Kitsune: an ensemble of autoencoders for online network intrusion detection," *arXiv preprint arXiv:1802.09089*, 2018.
- [151] D. Han, Z. Wang, Y. Zhong, W. Chen, J. Yang, S. Lu, X. Shi, and X. Yin, "Practical traffic-space adversarial attacks on learning-based nids. arxiv," 2020.
- [152] R. Abou Khamis, M. O. Shafiq, and A. Matrawy, "Investigating resistance of deep learning-based ids against adversaries using min-max optimization," in *ICC 2020-2020 IEEE International Conference on Communications (ICC)*, pp. 1–7, IEEE, 2020.
- [153] M. Usama, M. Asim, S. Latif, J. Qadir, et al., "Generative adversarial networks for launching and thwarting adversarial attacks on network intrusion detection systems," in *2019 15th international wireless communications & mobile computing conference (IWCMC)*, pp. 78–83, IEEE, 2019.
- [154] C. Benzaïd, M. Boukhalfa, and T. Taleb, "Robust self-protection against application-layer (D) DoS attacks in SDN environment," in *2020 IEEE Wireless Communications and Networking Conference (WCNC)*, pp. 1–6, IEEE, 2020.
- [155] M. J. Hashemi and E. Keller, "Enhancing robustness against adversarial examples in network intrusion detection systems," in *2020 IEEE Conference on Network Function Virtualization and Software Defined Networks (NFV-SDN)*, pp. 37–43, IEEE, 2020.
- [156] C. Zhang, X. Costa-Pérez, and P. Patras, "Tiki-Taka: Attacking and defending deep learning-based intrusion detection systems," in *Proceedings of the 2020 ACM SIGSAC Conference on Cloud Computing Security Workshop*, pp. 27–39, 2020.
- [157] R. Abou Khamis and A. Matrawy, "Evaluation of adversarial training on different types of neural networks in deep learning-based IDS," in *2020 international symposium on networks, computers and communications (ISNCC)*, pp. 1–6, IEEE, 2020.
- [158] J. Wang, J. Pan, I. AlQerm, and Y. Liu, "Def-IDS: An ensemble defense mechanism against adversarial attacks for deep learning-based network intrusion detection," in *2021 International Conference on Computer Communications and Networks (ICCCN)*, pp. 1–9, IEEE, 2021.
- [159] M. P. Novaes, L. F. Carvalho, J. Lloret, and M. L. Proença Jr, "Adversarial deep learning approach detection and defense against DDoS attacks in SDN environments," *Future Generation Computer Systems*, vol. 125, pp. 156–167, 2021.
- [160] Y. Wang, S. Lv, J. Liu, X. Chang, and J. Wang, "On the combination of data augmentation method and gated convolution model for building effective and robust intrusion detection," *Cybersecurity*, vol. 3, pp. 1–12, 2020.
- [161] H. Zhang, X. Yu, P. Ren, C. Luo, and G. Min, "Deep adversarial learning in intrusion detection: A data augmentation enhanced framework," *arXiv preprint arXiv:1901.07949*, 2019.
- [162] H. Larijani, N. Mtetwa, M. Yousefi, A. Javed, et al., "An adversarial attack detection paradigm with swarm optimization," in *2020 International Joint Conference on Neural Networks (IJCNN)*, pp. 1–7, IEEE, 2020.

- [163] I. Debicha, T. Debatty, J.-M. Dricot, and W. Mees, "Adversarial training for deep learning-based intrusion detection systems," arXiv preprint arXiv:2104.09852, 2021.
- [164] E. Anthi, L. Williams, M. Rhode, P. Burnap, and A. Wedgbury, "Adversarial attacks on machine learning cybersecurity defences in industrial control systems," *Journal of Information Security and Applications*, vol. 58, p. 102717, 2021.
- [165] N. Martins, J. M. Cruz, T. Cruz, and P. H. Abreu, "Analyzing the footprint of classifiers in adversarial denial of service contexts," in *Progress in Artificial Intelligence: 19th EPIA Conference on Artificial Intelligence, EPIA 2019, Vila Real, Portugal, September 3–6, 2019, Proceedings, Part II 19*, pp. 256–267, Springer, 2019.
- [166] E. Anthi, L. Williams, A. Javed, and P. Burnap, "Hardening machine learning denial of service (DoS) defences against adversarial attacks in IoT smart home networks," *computers & security*, vol. 108, p. 102352, 2021.
- [167] F. O. Catak and S. Y. Yayilgan, "Deep neural network based malicious network activity detection under adversarial machine learning attacks," in *Intelligent Technologies and Applications: Third International Conference, INTAP 2020, Grimstad, Norway, September 28–30, 2020, Revised Selected Papers 3*, pp. 280–291, Springer, 2021.
- [168] G. Apruzzese, M. Andreolini, M. Colajanni, and M. Marchetti, "Hardening random forest cyber detectors against adversarial attacks," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 4, no. 4, pp. 427–439, 2020.
- [169] A. McCarthy, P. Andriotis, E. Ghadafi, and P. Legg, "Feature vulnerability and robustness assessment against adversarial machine learning attacks," in *2021 International Conference on Cyber Situational Awareness, Data Analytics and Assessment (CyberSA)*, pp. 1–8, IEEE, 2021.
- [170] G. Apruzzese, M. Andreolini, M. Marchetti, V. G. Colacino, and G. Russo, "AppCon: Mitigating evasion attacks to ML cyber detectors," *Symmetry*, vol. 12, no. 4, p. 653, 2020.
- [171] I. Debicha, T. Debatty, J.-M. Dricot, W. Mees, and T. Kenaza, "Detect & reject for transferability of black-box adversarial attacks against network intrusion detection systems," in *International Conference on Advances in Cyber Security*, pp. 329–339, Springer, 2021.
- [172] M. Pawlicki, M. Choraś, and R. Kozik, "Defending network intrusion detection systems against adversarial evasion attacks," *Future Generation Computer Systems*, vol. 110, pp. 148–154, 2020.
- [173] H. Jiang, J. Lin, and H. Kang, "FGMD: A robust detector against adversarial attacks in the IoT network," *Future Generation Computer Systems*, vol. 132, pp. 194–210, 2022.
- [174] G. Apruzzese, M. Colajanni, L. Ferretti, and M. Marchetti, "Addressing adversarial attacks against security systems based on machine learning," in *2019 11th international conference on cyber conflict (CyCon)*, vol. 900, pp. 1–18, IEEE, 2019.
- [175] N. Wang, Y. Chen, Y. Xiao, Y. Hu, W. Lou, and Y. T. Hou, "MANDA: On adversarial example detection for network intrusion detection system," *IEEE Transactions on Dependable and Secure Computing*, vol. 20, no. 2, pp. 1139–1153, 2022.
- [176] A. Abusnaina, A. Khoramali, D. Nyang, M. Yuksel, and A. Mohaisen, "Examining the robustness of learning-based DDoS detection in software defined networks," in *2019 IEEE Conference on Dependable and Secure Computing (DSC)*, pp. 1–8, IEEE, 2019.
- [177] R. Chauhan and S. S. Heydari, "Polymorphic adversarial DDoS attack on IDS using GAN," in *2020 International Symposium on Networks, Computers and Communications (ISNCC)*, pp. 1–6, IEEE, 2020.
- [178] X. Fu, N. Zhou, L. Jiao, H. Li, and J. Zhang, "The robust deep learning-based schemes for intrusion detection in Internet of Things environments," *Annals of Telecommunications*, vol. 76, no. 5–6, pp. 273–285, 2021.
- [179] B. Nugraha, N. Kulkarni, and A. Gopikrishnan, "Detecting adversarial DDoS attacks in software-defined networking using deep learning techniques and adversarial training," in *2021 IEEE International Conference on Cyber Security and Resilience (CSR)*, pp. 448–454, IEEE, 2021.
- [180] A. Ganesan and K. Sarac, "Mitigating evasion attacks on machine learning based NIDS systems in SDN," in *2021 IEEE 7th International Conference on Network Softwarization (NetSoft)*, pp. 268–272, IEEE, 2021.
- [181] A. U. H. Qureshi, H. Larijani, M. Yousefi, A. Adeel, and N. Mtetwa, "An adversarial approach for intrusion detection systems using Jacobian saliency map attacks (JSMA) algorithm," *Computers*, vol. 9, no. 3, p. 58, 2020.
- [182] Y. Peng, G. Fu, Y. Luo, J. Hu, B. Li, and Q. Yan, "Detecting adversarial examples for network intrusion detection system with GAN," in *2020 IEEE 11th International Conference on Software Engineering and Service Science (ICSESS)*, pp. 6–10, IEEE, 2020.
- [183] S. M. Kasongo and Y. Sun, "Performance analysis of intrusion detection systems using a feature selection method on the UNSW-NB15 dataset," *Journal of Big Data*, vol. 7, pp. 1–20, 2020.
- [184] M. M. Alani, "Implementation-oriented feature selection in UNSW-NB15 intrusion detection dataset," in *International Conference on Intelligent Systems Design and Applications*, pp. 548–558, Springer, 2021.
- [185] V. Kumar, A. K. Das, and D. Sinha, "Statistical analysis of the UNSW-NB15 dataset for intrusion detection," in *Computational Intelligence in Pattern Recognition: Proceedings of CIPR 2019*, pp. 279–294, Springer, 2020.
- [186] A. Divekar, M. Parekh, V. Savla, R. Mishra, and M. Shirole, "Benchmarking datasets for anomaly-based network intrusion detection: KDD CUP 99 alternatives," in *2018 IEEE 3rd international conference on computing, communication and security (ICCCS)*, pp. 1–8, IEEE, 2018.
- [187] K. Vamsi Krishna, K. Swathi, P. Rama Koteswara Rao, and B. Basaveswara Rao, "A detailed analysis of the CIDDS-001 and CICIDS-2017 datasets," in *Pervasive Computing and Social Networking: Proceedings of ICPCSN 2021*, pp. 619–638, Springer, 2022.
- [188] S. Songma, T. Sathuphan, and T. Pamutha, "Optimizing intrusion detection systems in three phases on the CSE-CIC-IDS-2018 dataset," *Computers*, vol. 12, no. 12, p. 245, 2023.
- [189] W. Xu, J. Jang-Jaccard, A. Singh, Y. Wei, and F. Sabrina, "Improving performance of autoencoder-based network anomaly detection on NSL-KDD dataset," *IEEE Access*, vol. 9, pp. 140136–140146, 2021.
- [190] S. S. Panwar and Y. Raiwani, "Performance analysis of NSL-KDD dataset using classification algorithms with different feature selection algorithms and supervised filter discretization," in *Intelligent Communication, Control and Devices: Proceedings of ICICCD 2018*, pp. 497–511, Springer, 2020.
- [191] M. C. P. Saheb, M. S. Yadav, S. Babu, J. J. Pujari, and J. B. Maddala, "A review of DDoS evaluation dataset: CICDDoS 2019 dataset," in *International Conference on Energy Systems, Drives and Automations*, pp. 389–397, Springer, 2021.
- [192] M. Sakhalai and M. Wielgosz, "Modern cybersecurity solution using supervised machine learning," arXiv preprint arXiv:2109.07593, 2021.
- [193] A. Guerra-Manzanares, J. Medina-Galindo, H. Bahsi, and S. Nömm, "Using MedBIoT dataset to build effective machine learning-based IoT botnet detection systems," in *International Conference on Information Systems Security and Privacy*, pp. 222–243, Springer, 2020.
- [194] A. Alsaedi, N. Moustafa, Z. Tari, A. Mahmood, and A. Anwar, "TON_IoT telemetry dataset: A new generation dataset of IoT and IIoT for data-driven intrusion detection systems," *Ieee Access*, vol. 8, pp. 165130–165150, 2020.
- [195] N. Koroniotis, N. Moustafa, E. Sitnikova, and B. Turnbull, "Towards the development of realistic botnet dataset in the Internet of Things for network forensic analytics: Bot-IoT dataset," *Future Generation Computer Systems*, vol. 100, pp. 779–796, 2019.
- [196] Y. Mirsky, T. Doitshman, Y. Elovici, and A. Shabtai, "Kitsune: an ensemble of autoencoders for online network intrusion detection," arXiv preprint arXiv:1802.09089, 2018.
- [197] I. Homoliak and P. Hanacek, "ASNM datasets: A collection of network traffic data for testing of adversarial classifiers and network intrusion detectors [Internet]," *IEEE Dataport*, 2019.
- [198] J. H. Ring IV, C. M. Van Oort, S. Durst, V. White, J. P. Near, and C. Skalka, "Methods for host-based intrusion detection with deep learning," *Digital Threats: Research and Practice (DTRAP)*, vol. 2, no. 4, pp. 1–29, 2021.
- [199] M. Ghurab, G. Gaphari, F. Alshami, R. Alshamy, and S. Othman, "A detailed analysis of benchmark datasets for network intrusion detection system," *Asian Journal of Research in Computer Science*, vol. 7, no. 4, pp. 14–33, 2021.
- [200] R. A. Disha and S. Waheed, "Performance analysis of machine learning models for intrusion detection system using Gini impurity-based weighted random forest (GIWRF) feature selection technique," *Cybersecurity*, vol. 5, no. 1, p. 1, 2022.
- [201] R. Lyu, M. He, Y. Zhang, L. Jin, and X. Wang, "Network intrusion detection based on an efficient neural architecture search," *Symmetry*, vol. 13, no. 08, p. 1453, 2021.
- [202] W. Haider, *Developing reliable anomaly detection system for critical hosts: A proactive defense paradigm*. PhD thesis, UNSW Sydney, 2018.

- [203] E. K. Viegas, A. O. Santin, and L. S. Oliveira, "Toward a reliable anomaly-based intrusion detection in real-world environments," *Computer Networks*, vol. 127, pp. 200–216, 2017.
- [204] J. Cao, L. Lin, R. Ma, H. Guan, M. Tian, and Y. Wang, "An efficient deep learning approach to IoT intrusion detection," *The Computer Journal*, vol. 65, no. 11, pp. 2870–2879, 2022.
- [205] A. E. Aminanto and M. E. Aminanto, "Deep learning models for intrusion detection in wi-fi networks: A literature survey," *Sustainable Architecture and Building Environment: Proceedings of ICSDems 2020*, pp. 115–121, 2022.
- [206] M. Gharaibeh and C. Papadopoulos, "DARPA-2009 intrusion detection dataset report," *Tech. Rep.*, 2014.
- [207] C. Ardi, C. Aubry, B. Kocloski, D. DeAngelis, A. Hussain, M. Troglia, and S. Schwab, "The DARPA SEARCHLIGHT dataset of application network traffic," in *Proceedings of the 15th Workshop on Cyber Security Experimentation and Test*, pp. 59–64, 2022.
- [208] C. Thomas, V. Sharma, and N. Balakrishnan, "Usefulness of DARPA dataset for intrusion detection system evaluation," in *Data Mining, Intrusion Detection, Information Assurance, and Data Networks Security 2008*, vol. 6973, pp. 164–171, SPIE, 2008.
- [209] D. Biesner, K. Cvejosi, and R. Sifa, "Combining variational autoencoders and transformer language models for improved password generation," in *Proceedings of the 17th International Conference on Availability, Reliability and Security*, pp. 1–6, 2022.
- [210] M. AlSabah, G. Oligeri, and R. Riley, "Your culture is in your password: An analysis of a demographically-diverse password dataset," *Computers & security*, vol. 77, pp. 427–441, 2018.
- [211] D. Biesner, K. Cvejosi, B. Georgiev, R. Sifa, and E. Krupicka, "Advances in password recovery using generative deep learning techniques," in *Artificial Neural Networks and Machine Learning–ICANN 2021: 30th International Conference on Artificial Neural Networks, Bratislava, Slovakia, September 14–17, 2021, Proceedings, Part III 30*, pp. 15–27, Springer, 2021.
- [212] B. S. Khater, A. W. Abdul Wahab, M. Y. I. Idris, M. A. Hussain, A. A. Ibrahim, M. A. Amin, and H. A. Shehadeh, "Classifier performance evaluation for lightweight ids using fog computing in IoT security," *Electronics*, vol. 10, no. 14, p. 1633, 2021.
- [213] F. M. Ramos and X. Wang, "A machine learning based approach to detect stealthy cobalt strike C&C activities from encrypted network traffic," in *International Conference on Machine Learning for Networking*, pp. 113–129, Springer, 2022.
- [214] F. Ö. Sönmez and B. Günel, "Evaluation of security information and event management systems for custom security visualization generation," in *2018 International Congress on Big Data, Deep Learning and Fighting Cyber Terrorism (IBIGDELFT)*, pp. 38–44, IEEE, 2018.
- [215] K. M. Kavanagh, O. Rochford, and T. Bussa, "Magic quadrant for security information and event management," *Gartner Group Research Note*, 2015.
- [216] A. Bindra, "Securing the power grid: Protecting smart grids and connected power systems from cyberattacks," *IEEE Power Electronics Magazine*, vol. 4, no. 3, pp. 20–27, 2017.
- [217] R. Hiran, N. Carlsson, and N. Shahmehri, "Crowd-based detection of routing anomalies on the Internet," in *2015 IEEE Conference on Communications and Network Security (CNS)*, pp. 388–396, IEEE, 2015.
- [218] K. Wong, C. Dillabaugh, N. Seddigh, and B. Nandy, "Enhancing suricata intrusion detection system for cyber security in SCADA networks," in *2017 IEEE 30th Canadian Conference on Electrical and Computer Engineering (CCECE)*, pp. 1–5, IEEE, 2017.
- [219] S. Haas, R. Sommer, and M. Fischer, "Zeek-Osquery: Host-network correlation for advanced monitoring and intrusion detection," in *ICT Systems Security and Privacy Protection: 35th IFIP TC 11 International Conference, SEC 2020, Maribor, Slovenia, September 21–23, 2020, Proceedings 35*, pp. 248–262, Springer, 2020.
- [220] R. B. Tallane and D. W. Chandra, "Implementation of intrusion detection system (IDS) using security onion," *Journal of Syntax Literate*, vol. 7, no. 10, 2022.
- [221] M.-J. Sule, M. Li, G. A. Taylor, and S. Furber, "Deploying trusted cloud computing for data intensive power system applications," in *2015 50th International Universities Power Engineering Conference (UPEC)*, pp. 1–5, IEEE, 2015.
- [222] G. Zhang and A. Evry, "Implement directive rules on SIEM by detecting mal-behaviors,"
- [223] A. Haakila, "Implementing security monitoring at small and medium sized businesses," 2022.
- [224] N. d. A. M. Schumacher, "Anomaly detection models for cloud-edge intrusion detection in customer networks," 2022.
- [225] M. Alkasassbeh and S. Al-Haj Baddar, "Intrusion detection systems: A state-of-the-art taxonomy and survey," *Arabian Journal for Science and Engineering*, vol. 48, no. 8, pp. 10021–10064, 2023.
- [226] J. Chandler and A. Wick, "Synthesizing intrusion detection system test data from open-source attack signatures," in *2023 IEEE Security and Privacy Workshops (SPW)*, pp. 198–208, IEEE, 2023.
- [227] A. Adeyemo, "Design of an intrusion detection system (IDS) and an intrusion prevention system (IPS) for the EIU cybersecurity laboratory," 2016.
- [228] K. Hughes, K. McLaughlin, and S. Sezer, "Dynamic countermeasure knowledge for intrusion response systems," in *2020 31st Irish Signals and Systems Conference (ISSC)*, pp. 1–6, IEEE, 2020.
- [229] Y.-F. Hsu and M. Matsuoka, "A deep reinforcement learning approach for anomaly network intrusion detection system," in *2020 IEEE 9th international conference on cloud networking (CloudNet)*, pp. 1–6, IEEE, 2020.
- [230] C.-Y. Ho, Y.-D. Lin, Y.-C. Lai, I.-W. Chen, F.-Y. Wang, and W.-H. Tai, "False positives and negatives from real traffic with intrusion detection/prevention systems," *International Journal of Future Computer and Communication*, vol. 1, no. 2, p. 87, 2012.
- [231] M. Maliszewski, S. Pristerjahn, and U. Boryczka, "DBSCAN algorithm as a means to protect the ATM systems," in *2018 Innovations in Intelligent Systems and Applications (INISTA)*, pp. 1–6, IEEE, 2018.
- [232] A. Cosson, A. K. Sikder, L. Babun, Z. B. Celik, P. McDaniel, and A. S. Uluagac, "Sentinel: A robust intrusion detection system for IoT networks using kernel-level system information," in *Proceedings of the International Conference on Internet-of-Things Design and Implementation*, pp. 53–66, 2021.
- [233] R. Arthi and S. Krishnaveni, "Design and development of IoT testbed with DDoS attack for cyber security research," in *2021 3rd International Conference on Signal Processing and Communication (ICPSC)*, pp. 586–590, IEEE, 2021.
- [234] Y. Wang, D. Gu, J. Xu, and H. Du, "Hacking risk analysis of web trojan in electric power system," in *2009 International Conference on Web Information Systems and Mining*, pp. 510–514, IEEE, 2009.
- [235] M. Catalin and A. Cristian, "An efficient method in pre-processing phase of mining suspicious web crawlers," in *2017 21st International Conference on System Theory, Control and Computing (ICSTCC)*, pp. 272–277, IEEE, 2017.
- [236] W. Haider, G. Creech, Y. Xie, and J. Hu, "Windows based data sets for evaluation of robustness of host based intrusion detection systems (IDS) to zero-day and stealth attacks," *Future Internet*, vol. 8, no. 3, p. 29, 2016.
- [237] M. H. Raj, A. A. Rahman, U. H. Akter, K. N. Riya, A. T. Nijhum, and R. M. Rahman, "IoT botnet detection using various one-class classifiers," *Vietnam Journal of Computer Science*, vol. 8, no. 02, pp. 291–310, 2021.
- [238] B. Mukherjee, L. T. Heberlein, and K. N. Levitt, "Network intrusion detection," *IEEE network*, vol. 8, no. 3, pp. 26–41, 1994.
- [239] S. Roschke, F. Cheng, and C. Meinel, "Intrusion detection in the cloud," in *2009 eighth IEEE international conference on dependable, autonomic and secure computing*, pp. 729–734, IEEE, 2009.
- [240] M.-I. Nicolae, M. Sinn, M. N. Tran, B. Buesser, A. Rawat, M. Wistuba, V. Zantedeschi, N. Baracaldo, B. Chen, H. Ludwig, et al., "Adversarial robustness toolbox v1.0.0," *arXiv preprint arXiv:1807.01069*, 2018.
- [241] K. Roshan, A. Zafar, and S. B. U. Haque, "A novel deep learning based model to defend network intrusion detection system against adversarial attacks," in *2023 10th International Conference on Computing for Sustainable Global Development (INDIACom)*, pp. 386–391, IEEE, 2023.
- [242] V. Sehwag, S. Wang, P. Mittal, and S. Jana, "Hydra: Pruning adversarially robust neural networks," *Advances in Neural Information Processing Systems*, vol. 33, pp. 19655–19666, 2020.
- [243] B. Tarchoun, Q. M. Alam, N. Abu-Ghazaleh, and I. Alouani, "Fool the Hydra: Adversarial attacks against multi-view object detection systems," *arXiv preprint arXiv:2312.00173*, 2023.
- [244] G. W. Ding, L. Wang, and X. Jin, "Advertorch v0.1: An adversarial robustness toolbox based on pytorch," *arXiv preprint arXiv:1902.07623*, 2019.
- [245] D. Goodman, H. Xin, W. Yang, W. Yuesheng, X. Junfeng, and Z. Huan, "Advbox: a toolbox to generate adversarial examples that fool neural networks," *arXiv preprint arXiv:2001.05574*, 2020.
- [246] N. Papernot, F. Faghri, N. Carlini, I. Goodfellow, R. Feinman, A. Kurakin, C. Xie, Y. Sharma, T. Brown, A. Roy, et al., "Technical report

- on the cleverhans v2. 1.0 adversarial examples library,” arXiv preprint arXiv:1610.00768, 2016.
- [247] S. Asha and P. Vinod, “Evaluation of adversarial machine learning tools for securing ai systems,” *Cluster Computing*, pp. 1–20, 2022.
- [248] X. Ling, S. Ji, J. Zou, J. Wang, C. Wu, B. Li, and T. Wang, “DEEPSEC: A uniform platform for security analysis of deep learning model,” in 2019 IEEE symposium on security and privacy (SP), pp. 673–690, IEEE, 2019.
- [249] J. Rauber, R. Zimmermann, M. Bethge, and W. Brendel, “Foolbox native: Fast adversarial attacks to benchmark the robustness of machine learning models in PyTorch, Tensorflow, and JAX,” *Journal of Open Source Software*, vol. 5, no. 53, p. 2607, 2020.
- [250] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, “Membership inference attacks against machine learning models,” in 2017 IEEE symposium on security and privacy (SP), pp. 3–18, IEEE, 2017.
- [251] B. Kulynych, J. Hayes, N. Samarin, and C. Troncoso, “Evading classifiers in discrete domains with provable optimality guarantees,” arXiv preprint arXiv:1810.10939, 2018.
- [252] H. V. Vo, D. H. Nguyen, T. T. Nguyen, H. N. Nguyen, and D. V. Nguyen, “Leveraging AI-driven realtime intrusion detection by using WGAN and XGBoost,” in *Proceedings of the 11th International Symposium on Information and Communication Technology*, pp. 208–215, 2022.
- [253] M. Kahng, N. Thorat, D. H. Chau, F. B. Viégas, and M. Wattenberg, “GAN lab: Understanding complex deep generative models using interactive visual experimentation,” *IEEE transactions on visualization and computer graphics*, vol. 25, no. 1, pp. 310–320, 2018.